

Received 5 December 2025, accepted 19 December 2025, date of publication 5 January 2026, date of current version 9 January 2026.

Digital Object Identifier 10.1109/ACCESS.2025.3650636

 SURVEY

Active Network Management Integration in Expansion Planning of Active Distribution Networks: A Comprehensive Review

IRINA LAAMANEN^{ID 1}, SAMI REPO^{ID 1}, AND THOMAS STETZ^{ID 2}

¹Unit of Electrical Engineering, Tampere University, 33720 Tampere, Finland

²Department of Electrical Engineering and Information Technology, THM University of Applied Sciences Mittelhessen, 35390 Giessen, Germany

Corresponding author: Irina Laamanen (irina.salazarfonseca@tuni.fi)

This work was supported in part by the Fortum and Neste Foundation under Grant 20230016, and in part by the Business Finland under the Project Energy Community as a Driver of Electrified City (ECADEC).

ABSTRACT This paper presents a comprehensive review of the recent advancements in active distribution network expansion planning, focusing on the integration of active network management technologies and services. The transition from passive to active distribution networks is driven by the proliferation of distributed energy resources such as renewable and non-renewable distributed generation, energy storage systems, and demand response, which necessitate new planning and operational paradigms. The active distribution network expansion planning models must account for the stochastic and dynamic nature of distributed energy resources, requiring multi-objective, multi-stage, and hierarchical optimization approaches. The paper analyzes over 50 articles published since 2014, categorizing them based on planning horizons, distributed energy resources integration, solution methods, uncertainty handling techniques, and reliability metrics. Active network management technologies such as dynamic thermal rating, reactive power control, and coordinated voltage control are explored in depth. The review highlights how these technologies enable more flexible and cost-effective grid expansion by accommodating more distributed energy resources and reducing infrastructure investment needs but also reveals recurring gaps related to multi-actor coordination, service-based flexibility procurement, and the omission of active network management implementation costs. This work serves as a vital resource for researchers and utility planners in selecting appropriate active network management strategies and optimization techniques for active distribution networks.

INDEX TERMS Active distribution networks, active network management, distributed energy resources, distribution network expansion planning.

I. INTRODUCTION

Grid modernization, expansion, and smartening are highly needed in both transmission and, increasingly, distribution levels to enable the energy transition across all economic sectors. Given the rising numbers in the last decades of decentralized renewable installations, the electrification of transport, heating and industry, and more active consumers, the role of distribution system operators (DSOs) has drastically changed from a formally passive distributor toward an active enabler and facilitator of the transition [1], [2].

The associate editor coordinating the review of this manuscript and approving it for publication was Peter Palensky^{ID}.

In this transition process, new technologies, operational patterns, and players have emerged (e.g., distributed energy resources (DERs), flexibility services, flexibility service providers (aggregators), local flexibility markets, and prosumers). The distributed energy resources can include technologies such as renewable and non-renewable distributed generation (DG), energy storage systems (ESSs), and demand response (DR) [3].

DERs are often positioned within the distribution networks and behind the metering point, which impacts power flow distribution and the power grid control and protection system [4]. On the other hand, such resources can provide flexibility services in the grid and different markets where end-users

have been seen as passive participants. In this context, the end-user gains a new role, being able to provide flexibility services for the DSOs [3].

However, the large proliferation of DERs in the existing passive distribution networks, designed for unidirectional power flows and centralized control, will bring implications for the operation, expansion planning, and design of the future grid. To fully capitalize on the potential of DERs while ensuring a reliable and efficient power supply, a paradigm shift towards active distribution networks (ADNs) is essential [5].

An active distribution network is defined as a distribution network comprising distributed or decentralized energy sources with real-time operation and control capabilities [4]. The ADN is a new solution to the flexible utilization of distributed energy resources to suit the characteristics of the distribution network [6]. In the ADNs, the DSOs can purchase the flexibility services offered by the network's end-user ("customers") through the flexibility service providers or sometimes directly from resources as mandated by the grid codes [7].

The ADN's operation, expansion planning, and design have become more complex and challenging in such a context. Consequently, the approaches used in passive distribution network expansion planning (e.g., the so-called "fit and forget" approach [8], [9], the worst-case principle [10], or "copper plate" approach [11], [12]) are not capable of providing an adequate solution for ADNs. This is because active elements of grid management (e.g., remotely controllable switches, reactive power control, auto-recloser, etc.) and the stochastic impacts of DERs not managed by DSO are not considered. Numerous other factors, such as the stochastic nature of electric vehicle (EV) charging demand, the dynamic behavior of customers responding to day-ahead market prices and participating in flexibility markets like balancing and reserve markets, and the intermittent nature of renewable DG (e.g. wind and photo-voltaic power plants) introduce compounded uncertainties into load and generation profiles. These fluctuations can lead to operational disruptions, including voltage quality problems, overloading of lines and transformers, and even cascading failures exacerbating the challenges faced by distribution system operators [5].

Unlike the conventional "fit and forget" approach, active network management (ANM) has emerged as a pivotal strategy in active distribution network expansion planning (ADNEP) to support the increasing penetration of DERs [13], [14], [15]. ANM is defined as the exploitation of flexible network assets to provide a secure means of increasing grid utilization [16]. At its core, an ANM solution consists of a control system integrated with communication technologies and the power system, capable of dynamically managing DERs and grid assets to serve the operational objectives of DSOs. By leveraging real-time control and coordination, ANM can reduce reliance on conventional grid reinforcements, accelerate DER connection processes, and ultimately lower infrastructure and operational costs for customers [16].

As illustrated in Fig. 1, the choice of distribution network planning methodology is closely dependent on the level of DER integration. When DER penetration is low, traditional expansion planning approaches can still offer cost-effective and technically viable solutions. However, as DER integration reaches moderate levels, the inclusion of ANM becomes essential, alongside the strategic use of network flexibility to maintain system reliability and efficiency. At high DER penetration levels, more advanced planning is required—not only incorporating ANM solutions but also introducing new market frameworks, such as local flexibility markets (LFMs), to effectively manage distributed flexibility and ensure coordinated operation [3], [14]. Consequently, the integration of ANM significantly influences ADNEP models, necessitating a paradigm shift in how distribution networks are planned and operated in the presence of high DER variability and uncertainty.

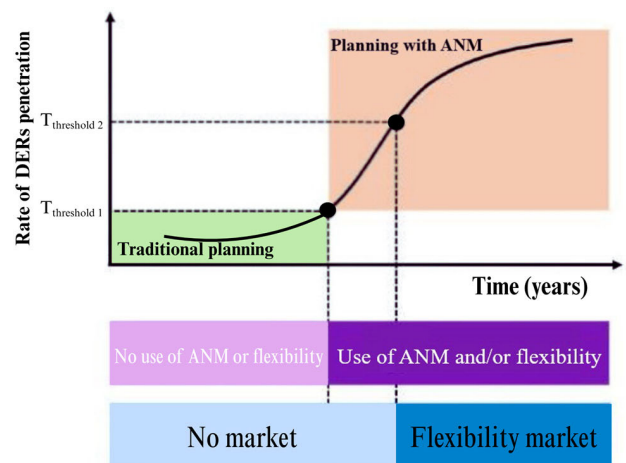


FIGURE 1. Evolution of the distribution network according to the development of DERs(modified from [14]).

Therefore, ADNEP models in terms of both objective functions and implementation methods have significantly changed [17], providing a holistic approach that allows the power distribution network to accommodate DERs, assess their value, and address grid uncertainties across distinct planning horizons. The current challenges in ADNs expansion solutions have encouraged many studies, showing that ADNEP models are complex optimization problems and challenging to solve in large-scale real power systems [18], [19], [20], [21], [22], [23]. Those studies include the ADNEP models that analyze DERs integration [24], [25], grid reliability aspects [18], [26], network efficiency improvement, hosting capacity enhancement, multi-objective ADNs expansion planning [27], [28], formation of microgrids (MGs), and ADNs management. As well as emerging research that includes dynamic customer responses to day-ahead and intraday, balancing and reserve markets, utilization of local flexibility markets, and utilization of flexible connection contracts [3], [29].

Despite the inherent complexity of ADNEP models, the incorporation of ANM technologies and services enables improved utilization of existing distribution feeders, increases the hosting capacity for renewable and non-renewable DG, and enhances the coordinated control of system elements. ANM can also influence distribution system reliability by improving reliability indices, reducing reinforcement needs, and lowering connection costs [30], [31], [32]. In addition, when DSOs procure flexibility services from DERs, ANM can mitigate adverse DG impacts such as bidirectional power flows, voltage deviations, and undesired islanding. These observations motivate the objective of this paper, which is to provide a comprehensive review of how ANM technologies and services are represented in ADNEP models. The review covers 50 studies published after 2014 and follows concepts and regulatory principles aligned with the European energy market.

Although several prior reviews have addressed important elements of distribution network planning including: general planning models, the emergence of active distribution systems, uncertainty modelling techniques, coordination mechanisms, and reliability considerations [5], [30], [32], [33], [34], [35], [36], [37] to the authors' knowledge, none have provided a detailed examination of how ANM technologies and services are formulated within ADNEP models. Earlier works by [34] and [35] describe the transition toward active distribution networks, while [36] focus on coordination aspects, yet these studies do not analyze the mathematical embedding and operational impact of ANM within expansion planning formulations. Similarly, reviews by [5], [30] concentrate on uncertainty treatment, [32] provide a reliability-oriented perspective, [33] offers a broad synthesis of distribution network expansion planning developments, and [37] examine transactive energy frameworks; however, none of these reviews explore how specific ANM actions such as reactive power control, coordinated voltage control, or network reconfiguration reshape ADNEP objectives, constraints, and planning decisions. In contrast, the present review contributes a technology-level and model-oriented synthesis, clarifying how ANM is implemented within modern ADNEP studies. By doing so, this work complements and extends the coverage of previous surveys by providing a systematic and detailed analysis of ANM integration, which is essential for DSOs seeking cost-effective, flexible and reliable expansion strategies.

The rest of the article is structured as follows: Section II examines the significant aspects of the active networks and expansion planning, along with key features of the ADNEP models and the summary of the reviewed reliability aspects in ADN expansion planning problem; Section III provides a detailed literature review and discussion of active network management technologies and services in the active distribution networks expansion planning models; discussion and future challenges are discussed in Section IV; and finally conclusions are presented in Section V.

II. THE EXPANSION PLANNING OF ACTIVE DISTRIBUTION NETWORKS: AN OVERVIEW

A. DATA SOURCES, INCLUSION, AND EXCLUSION CRITERIA

The data sources for this literature review include a range of academic databases and repositories to ensure a wide coverage of relevant literature. The primary databases consulted were IEEE Xplore, ScienceDirect, and Google Scholar. These platforms were chosen due to their extensive collections of scholarly articles, conference proceedings, and technical reports in the fields of engineering and technology, among others.

The search involved using specific keywords and phrases related to the research topic. Keywords included "active distribution network," "expansion planning," "active network management," "distributed energy resources," and "demand response." Boolean operators such as "AND" and "OR" were employed to combine terms and refine search results. The search was limited to publications from 2014 to 2025 to capture the most recent developments and trends in the field. With these search criteria, 190 articles were initially selected, and after reading the abstract and conclusions only 50 papers will be analyzed. Exclusion criteria involved omitting studies that were not directly related to the active distribution network expansion planning. Articles that did not include DERs or those that focused on unrelated sectors were excluded. Additionally, this review did not consider non-peer-reviewed sources, opinion pieces, and publications in languages other than English.

B. DEFINITION OF ACTIVE DISTRIBUTION NETWORKS

Traditional passive distribution networks were designed for predictable, unidirectional power flows from centralized generation toward consumers, with limited observability, minimal automation, and deterministic planning principles [34], [38]. Within this paradigm, customers and DERs functioned essentially as non-active system elements, and operational decisions centered on maintaining basic voltage and thermal limits using fit-and-forget reinforcement strategies [39], [40]. However, rapid growth in renewable generation, electric vehicles, flexible loads, and energy communities has introduced variability, bidirectional flows, and new operational uncertainties that challenge the adequacy of conventional approaches [17], [31].

In response, modern power systems are evolving toward active distribution networks. CIGRE C6.19 developed a shared global definition of an active distribution network. The first definition of ADNs was presented in [41] in the year 2009 as "*Active distribution networks are distribution networks that have systems in place to control a combination of distributed energy resources. Distribution system operators have the possibility of managing the electricity flows using a flexible network topology. DERs take some degree of responsibility for system support, which will depend on a*

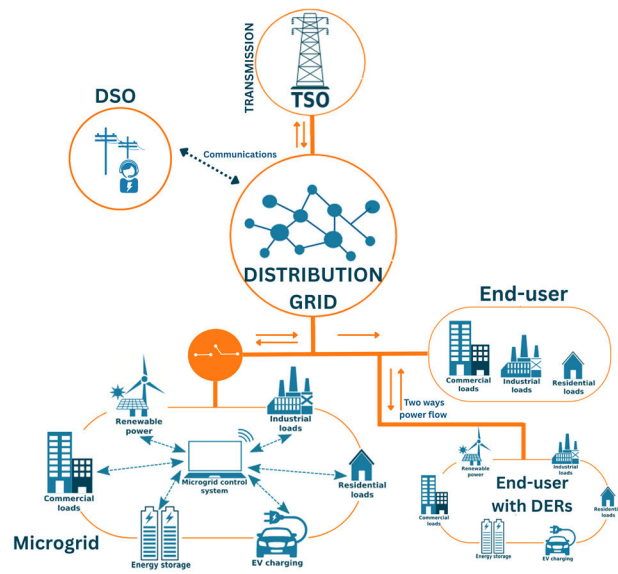


FIGURE 2. Component-level representation of the active distribution networks.

suitable regulatory environment and connection agreement". Fig. 2 illustrates the component-level presentation of ADNs considering domains.

The concept of ADNs encompasses various notions characterized by pervasive sensing, advanced communication, automation, and real-time coordination of DER, storage, and flexible demand resources. ADNs operate as cyber-physical energy systems, integrating multi-level data flows, digital twins, and distributed decision-making frameworks that leverage flexibility services to support voltage control, congestion management, resilience, and market participation [42], [43], [44]. This shift is enabled by bi-directional communication networks linking home devices, neighborhood-level systems, field equipment, and utility-wide control platforms, along with standardized data models and interoperable protocols [38], [45]. At the same time, energy sector coupling linking electricity with heating, gas, and mobility infrastructures demands coordinated planning approaches and multi-stakeholder data exchange that extend far beyond the traditional electricity silo [17], [46].

These developments fundamentally reshape planning and operational processes. Modern ADN planning incorporates DERs optimization, uncertainty modelling, flexibility evaluation, and multi-market coordination using probabilistic, robust, and hybrid methods [3], [5], [30]. Operationally, the distribution system operator transitions from a passive network operator to an active system manager capable of procuring and coordinating flexibility, enabling microgrid operation, and interacting with transmission system operators (TSOs) and local energy markets [4], [46], [47]. Several efforts have been made to develop different models for ADNPE, which consider not only the planning scheme but also the operation of the network [19], [20], [48]. The

following subsection examines the significant aspects of the expansion planning, along with key features of the ADNPE models.

C. FEATURES OF ADNS EXPANSION PLANNING

The traditional distribution network expansion planning (DNEP) problem aims to determine, in the most economical manner, the investments required in the distribution system to satisfy increasing demand, maintain secure operation, and ensure adequate quality of service to consumers. The formulation encompasses a broad set of decision variables that include *integer variables* associated with the size, type, and location of new or reinforced infrastructure, such as substations, transformers, feeders, capacitors, and switches, as well as *continuous variables* that characterize system operation, including DG power injection, bus voltage magnitudes, active and reactive power flows, ESSs charging and discharging power, and potential load shedding. The coexistence of discrete investment variables and continuous operational variables, combined with the non-linear equations governing power flow and network performance, results in a mixed-integer non-linear programming model. Solving this model constitutes a highly complex task due to its combinatorial structure and the significant number of technical, financial, and operational constraints involved [3].

Nevertheless, ADNPE shares the main features of DNEP but has evolved with new developments and features, as shown in Fig.3. Modern ADNPE considers the effects from a range of active network interventions that are exercised at differing time scales while capturing the intrinsic stochastic behavior of renewable generation or demand response

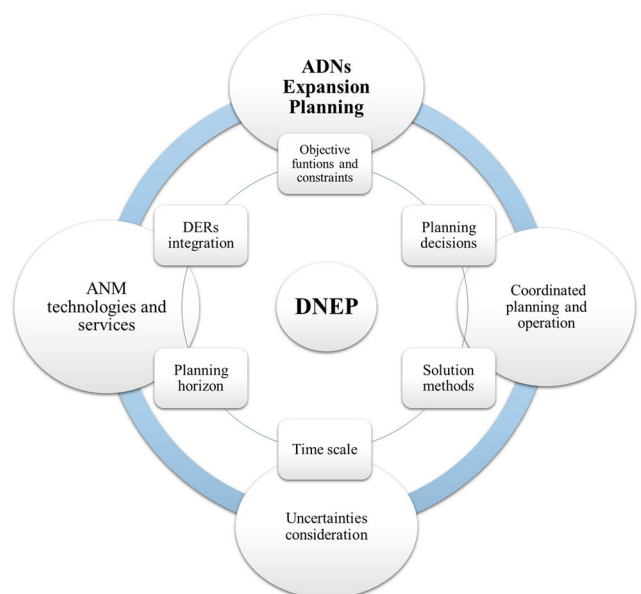


FIGURE 3. Features of the active distribution network expansion planning.

to produce strategic plans that are robust against a highly uncertain energy future [6].

The principal features of the ADNEP models are generally defined as the planning horizon, the time scale for planning, planning decisions, DERs integration, optimization objectives and constraints, ANM technologies and services, and solution methods [3], [13], [30], [33], [38]. The ADNEP models explicitly consider various uncertain input parameters (i.e., intermittent renewable generation, load demand, and electricity market prices) [3]. As well as the use of multi-objective planning techniques to sort out viable solutions among conflicting objectives that satisfy multiple stakeholders [49], and coordinated planning and operation [6], [13]. In the following subsections, these features are discussed in detail.

The first three features to be defined in the ADNEP models are the planning horizons, time scale, and planning decisions. The typical planning horizons are: (i) short-term (planning horizon of up to one year) [30]; (ii) mid-term planning horizon during the next 5 to 10 years where DSOs across Europe are required to prepare distribution network development plans outlining their investment needs [50]; and (iii) long-term planning (>10 years) strategic infrastructure and flexibility planning [51], [52], see Fig 4. In the short-term planning, the main targets are connecting new customers to the existing grid, enhancing reliability in specific areas, enhancing hosting capacity [53]. ANM measures are used to avoid or defer capital reinforcement while enabling near-term connections. Meanwhile, in the mid-term planning, extending the existing grid's lifetime to host more DERs and enhancing reliability are the principal targets.

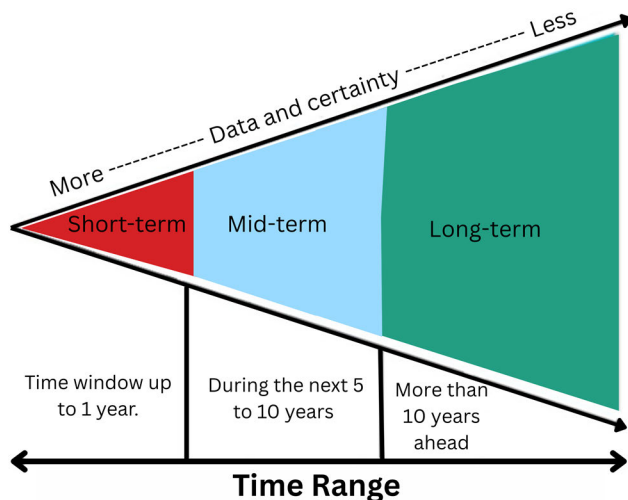


FIGURE 4. Time cone for planning horizons.

Long-term planning studies cover future demand growth, technology, or policy targets [54]. They deal with several uncertainties, the upgradation of existing infrastructure, installation of new capacity [55], and expansion of the grid to new areas. Long-term planning adopts scenario-based,

strategic network directions developed in coordination with land-use and regional planning authorities, valuing flexibility options.

The time scale for planning can be categorized: (i) single-stage is also called static [24], [56], [57], which determines the planning requirements in only one stage; or (ii) multistage, which is also called dynamic [58], [59]. This type determines the planning requirements for successive expansion plans over several planning stages. More useful results are offered by the dynamic approach, which is, however, more challenging because of the interdependency between stages [34].

From the planner perspective there are several decisions that can be implemented in the solution of the expansion planning of the grid. Those planning decisions are related to when/where, how many and what type of substation, feeders and other electrical components need to be installed or upgrade in the grid [30], [33], [38]. It is worth to notice that in the European countries the size, type, location and installation time of energy storage facilities are determined by the owners [60]. Furthermore, DG owners make decisions on the location, size and the type of DGs to install based on geographical conditions, fuel resources, and feasibility of using wind turbines, photovoltaic panels, biomass, etc [38].

Network reconfiguration is another planning decision used in the ADNEP models. In [18] and [51] the authors applied network reconfiguration and line reinforcement for the stochastic expansion planning model proposed in their articles. Frequently, the planning decisions are combined; one of the typical combinations is the size and location of DGs, capacitor banks, and ESSs, as used in the work presented in [19], [61], [62], and [63].

1) DISTRIBUTED ENERGY RESOURCES

One of the major drivers of the transition from passive to active DN is the massive integration of DERs. These resources can provide flexibility services (e.g., frequency control and balancing, congestion management, voltage control, and security of supply/islanded operation) for DSOs and TSOs. DSOs can purchase flexibility in several ways (e.g., connection agreement, network tariffs, and market) [64]. However, to ensure maximum availability of flexibilities for different local and system-level services, ANM principles are needed to avoid local flexibility use restrictions in distribution networks due to congestion related to thermal or voltage limit violations [65].

In today's terminology, a DER can refer to various types of resources with different characteristics that implement vastly different technologies. All types of DG (i.e., solar panels, wind turbines, small hydropower plants, combined heat and power (CHP) units, etc.) are DERs. Furthermore, different energy storage systems (such as batteries or flywheels) are also considered DERs. Finally, demand responses like optimized electric vehicle charging or heating/cooling are

deemed to be DERs today [66]. A brief explanation of these resources is given below.

Renewable and non-renewable DG units are small generating sources (5 kW to 10 MW) connected directly to the distribution system or behind the consumer meter [67], [68]. The distributed generation may offer, in the most optimal case, several advantages, including environmental benefits, reduction of on-peak network operation cost, deferral of distribution network reinforcements, lower power and energy losses, lower transmission and distribution costs, and possible increase of the reliability of the distribution network [69], [70]. Nevertheless, the integration of DGs poses serious challenges to DSOs, because renewable DG units are variable (not dispatchable) generation sources [67]. On the other hand, DGs and ESSs are owned and operated by third parties creating security and power quality challenges.

The energy storage system is defined as a group of equipment that allows the storage of several forms of energy to be drawn upon at a later time to perform useful operations [53]. ESS is so interesting and valuable to power systems because it can act either as a load or as a generator, with increased advantages when the battery storage system is connected to the grid through power electronic inverters. Moreover, ESSs help increase the penetration of renewable DG by realizing production following and providing flexibility services like congestion management and island operation, which may help to increase the energy efficiency of the complete energy system and reduce the electricity market price or increase the value of renewable electricity on the market by providing energy arbitrage services [67].

Demand response means the change of electricity load by final customers from their normal or current consumption patterns in response to market or other signals, including in response to time-variable electricity prices or incentive payments or in response to the acceptance of the final customer's bid to sell demand reduction or increase at a price in an organized market [60]. Demand response involves load shifting, valley filling, or peak clipping to provide flexibility in different markets and grid management. It is based on two main mechanisms: indirect price-based mechanisms (or implicit demand response), which use price signals and tariffs to incentivize consumers to shift consumption to cheaper periods, and direct control mechanisms (or explicit demand response), which monetize flexibility through direct payments to consumers who have an obligation to provide flexibility via a bilateral contract like a flexible grid connection contract, or trading at flexibility market [3].

2) ACTIVE NETWORK MANAGEMENT

Active network management is a core part of the concept of active distribution networks and is one of the principal features of ADNEP. Active network management's vital role is managing network constraints arising from general load variation, excess renewable generation, or DER variability by combining traditional network assets and DER

flexibility [66]. ANM uses flexible assets to maintain safe and reliable operation despite high network loading. At the core of ANM is control of active and reactive power to maintain acceptable voltages throughout the network as well as avoiding overloads [16].

Development of ANM is however not new and has been the focus of several recent activities, such as: Active Response, ANM4L (Active Network Management for All), ADINE (Active DIstribution NETwork) and IDE4L (Ideal Grid for All) projects [9], [16], [71], [72]. Many of these projects are driven by DSOs, TSOs, local initiatives and communities, highlighting the importance of implementing innovative solutions into the operation and planning of the power system. Furthermore, several publications have investigated the implementation of ANM technologies and services in the ADNEP models [23], [48], [62], [73], [74]. Providing results that show ANM solutions enables a move from the conservative 'fit-and-forget' approach to an actively managed network that makes best use of the available capacity [15].

The ANM strategies are implemented in the short and mid-term planning horizon as a key for cost-effective integration of DERs to postpone the grid investments for increase the hosting capacity [15], [23] and to address congestion and voltage problems [53]. Active network management technologies and services include monitoring, protection and control of the ADN. The main ANM solutions considered in the literature are shown graphically in Fig. 5.

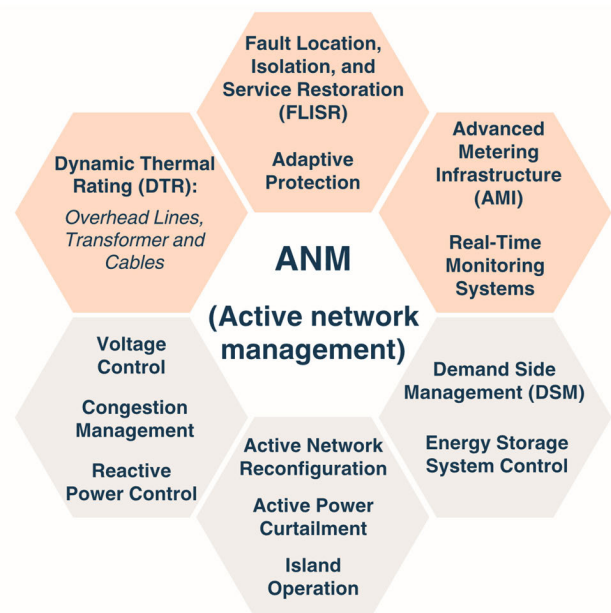


FIGURE 5. Typical active network management technologies and services for ADNs.

From the ANM technologies and services illustrated in Fig. 5, some of them, such as: reactive power control (RPC), active power curtailment, coordinated voltage control, network reconfiguration, and DTR have been successfully implemented in the reviewed publications [18], [23], [59],

[75], [76], [77]. Section III provides a detailed summary of the articles that included ANM technologies and services in the ADNEP problem.

3) OBJECTIVE FUNCTIONS AND CONSTRAINTS

Any expansion planning process aims towards achieving maximization and/or minimization of certain objectives, as shown in Fig. 6, that can be related to technical indices, investment and operational costs of the ADN [30], [34], [49], [78]. Most of the adopted objective functions aim to minimize the annualized net present value (NPV) of costs in terms of investment (fixed costs) and operation (variable costs) [3], [33]. The efficacy and practical relevance of the solution of any technical problem depends not only upon the effective formulation of an objective function but also on the choice of constraints, which limits the solution space [79].

The most common technical constraints include (a) power balance equations, (b) Feeder capacity limits, (c) substation capacity limits, (d) renewable and non-renewable DG nominal capacity limits, (e) radial topology, (f) voltage limits, (g) reliability indexes, (h) operational constraints of ESSs, (i) operational constraints of DR, (j) position limits on-load tap changer (OLTCs), and (k) power factor [13], [26], [79].

The economic constraints mainly refer to the budget limits for DSOs to build substations and feeders and budget limits for distributed generation operators (DGOs) to install DGs. Additionally, some articles, such as [61], introduced the constraint of limiting the total annual CO² emissions from the distribution system to a certain value.

Investment Costs	Operational Costs	Technical Indices	Others
• Installing (e.g., transformers, substation, feeders, and electrical components) • Network reinforcement	• Energy losses • Expected energy not supplied • Demand response • Active network management	• Power supply quality • Voltage deviation • Power losses • System reliability	• Social welfare • Renewable energy sources penetration • CO₂ emissions

FIGURE 6. Objectives of the active distribution network expansion planning.

From an optimization perspective in the articles reviewed, three types of formulation were used to solve the ADNEP problem: (I) single-objective problem, (II) multiobjective problem, and (III) multilevel optimization.

Single-objective optimization refers to an optimization problem where the goal is to optimize (maximize or minimize) a single objective function ($f(\mathbf{x}')$) while considering a set of constraints ($g(\mathbf{x}')$) [80]. Generally, a single-objective problem can be defined as follows:

$$\begin{aligned} & \text{Minimize } f(\mathbf{x}') \\ & \text{Subject to } g_j(\mathbf{x}') \geq 0, \quad j = 1, 2, \dots, m \\ & \quad \mathbf{x}' \in X \subset R^n \end{aligned} \quad (1)$$

where $\mathbf{x}'$ is a vector of n decision variables, $\mathbf{x}' = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)^T$, and X represents a feasible region.

Multi-objective optimization refers to finding the optimal solution values of more than one desired goal. Generally, a multi-objective problem that has several conflicting objectives which are to be optimized simultaneously and mathematically, can be formulated as follows [81], [82]:

$$\begin{aligned} & \text{Minimize } f_i(\mathbf{x}) = \{f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_k(\mathbf{x})\} \\ & \text{Subject to } \mathbf{x} \in S \subset R^n \end{aligned} \quad (2)$$

where $f_i(\mathbf{x})$ for $i = 1, 2, \dots, k$ are objectives to be minimized. The decision vector $\mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)^T$ belongs to the feasible region S . The objective vector $\mathbf{z} = \mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), \dots, f_k(\mathbf{x}))^T$ belongs to the objective space R^n .

The multiobjective formulations can be classified as:

- Scalarization method so-called multiobjective with weights. In this approach the multiple objectives are transformed into a single objective thanks to the user-defined fixed weights of the individual objectives and the planner determines the relative importance of the objective functions before executing the optimization algorithm [34], [56], [83].
- Pareto method: in this approach, the goal is to find a set of solutions (Pareto frontier) that represent the best trade-offs between objectives, which are Pareto optimal set solutions. Solutions on the Pareto front represent compromises between the different objectives, and decision-makers can choose one solution based on preference [18], [28], [82].

Multilevel optimization is defined as mathematical programming developed to address compromises between the interactive decision entities distributed throughout a hierarchical organization. In a multilevel decision-making process, decision entities at the upper level and the lower level are termed the leader and the follower and make their individual decisions in sequence to optimize their respective objectives. This decision-making process means that the leader has priority in making his own decision, and the follower reacts after and in full knowledge of the leader's decision; however, the leader's decision is implicitly affected by the follower's reaction [84], [85]. Two typical multilevel decision forms have been used in the expansion planning problem: bi-level and tri-level optimization. The general formulation for bi-level optimization [86], [87] is described as:

$$\begin{aligned} & \min_{\mathbf{x} \in X, \mathbf{y}} F(\mathbf{x}, \mathbf{y}) \\ & \text{Subject to } G(\mathbf{x}, \mathbf{y}) \geq 0, \\ & \quad \mathbf{y} \in S(\mathbf{x}) \end{aligned} \quad (3)$$

where $S(\mathbf{x})$ is the set of optimal solutions of the $\mathbf{x}$ -parameterized problem

$$\begin{aligned} & \min_{\mathbf{y} \in Y} f(\mathbf{x}, \mathbf{y}) \\ & \text{Subject to } g(\mathbf{x}, \mathbf{y}) \geq 0 \end{aligned} \quad (4)$$

Equation (3) is the upper-level objective function $F: R^n \times R^m \rightarrow R$ and (4), is the lower-level objective

function $f : R^n \times R^m \rightarrow R$. Where $G : R^{n_x} \times R^{n_y} \rightarrow R^m$, $g : R^{n_x} \times R^{n_y} \rightarrow R^l$ are the constraint conditions of the upper-level and the lower-level respectively. Moreover, $x \in R^{n_x}$ and $y \in R^{n_y}$ are the upper-level and lower-level variables respectively [86].

A bi-level problem contains two levels of optimization tasks where one optimization task is nested within the other. The outer optimization problem is commonly referred to as the leader's (upper level) optimization problem and the inner optimization problem is known as the follower's (or lower level) optimization problem. The two levels have their own objectives and constraints as illustrated in (3) and (4) [86], [87].

In the specific context of ADNEP, the leader–follower hierarchy almost always reflects a separation between long-term infrastructure planning and short-term system operation or market behavior. In several bi-level formulations, the upper level represents the DSO or distribution company, which decides on network reinforcements, siting and sizing of DG and ESS, and sometimes on tariff signals or contractual terms. The lower level then solves a detailed operational problem (optimal power flow or market clearing) that reacts optimally to the upper-level decisions.

For instance, the robust bi-level model in [73] places the DSO at the upper level, deciding on the investment in OLTCs, ESS, capacitor banks and static var compensators, while the lower level minimizes the worst-case operating cost under an adjustable uncertainty set. In this setting, ANM variables such as OLTC taps, ESS schedules and reactive power injections are embedded in the follower's problem and influence the upper-level cost through reduced losses, curtailed energy and load shedding. A similar planning–operation decomposition is used in [88], where the upper level maximizes the total profit of an integrated community energy system (ICES) by deciding on PV and ESS investments, while the lower level represents prosumer-level operation and utility, capturing how active customers react to the ICES investment and contract structure.

Other bi-level ADNEP formulations explicitly model conflicting objectives between the DSO and DER owners. In [89], the upper level simultaneously minimizes total system costs and EENS while maximizing renewable DG penetration, whereas the lower level maximizes the net profit of DG owners and the reserve capability of ESSs. The leader's investment and tariff decisions define the feasible region and profitability of DER portfolios, while the follower's optimal operation feeds back into the upper-level cost and reliability indices. Furthermore, in [90] the upper level represents a non-monopolistic distribution stakeholder organization that minimizes the total cost imposed on all actors, while the lower level minimizes generation costs given the expansion plan, effectively embedding a market-clearing optimal power flow inside the planning problem.

On the other hand, in a tri-level hierarchical decision problem, each decision entity at one level has its objective,

constraints, and decision variables that are affected in part by the decision entities at the other two levels [84], a general tri-level decision problem is defined as:

$$\begin{aligned} \min_{x \in X} f_1(x, y, z) \\ \text{Subject to } g_1(x, y, z) \leq 0, \\ x \in X \subset R^p, y \in Y \subset R^q, z \in Z \subset R^r \end{aligned} \quad (5)$$

where, for each x given by the first level (5), (y, z) solve the problems of the second and third levels (6) and (7):

$$\begin{aligned} \min_{y \in Y} f_2(x, y, z) \\ \text{Subject to } g_2(x, y, z) \leq 0 \end{aligned} \quad (6)$$

where, for each (x, y) given by the first and second levels, z solves

$$\begin{aligned} \min_{z \in Z} f_3(x, y, z) \\ \text{Subject to } g_3(x, y, z) \leq 0 \end{aligned} \quad (7)$$

where x, y, z are the decision variables and $f_1, f_2, f_3 : R^p \times R^q \times R^r \rightarrow R^1$ are the objective functions of the three levels respectively; $g_i : R^p \times R^q \times R^r \rightarrow R^{k_i}, i = 1, 2, 3$ are the constraint conditions of the three levels respectively.

While many studies on multilevel decision-making have focused on bi-level decision-making, research on tri-level decision-making problems has increasingly attracted investigations into solution approaches since tri-level decision-making can be applied to handle many decentralized decision problems in the real world [84]. In the ADNEP context, tri-level models extend this hierarchy by separating network expansion, DER siting/sizing, and operational dispatch into distinct decision layers.

In [91], the upper level determines network expansion and reinforcement to minimize investment, operation and maintenance costs and EENS while maximizing RES penetration;

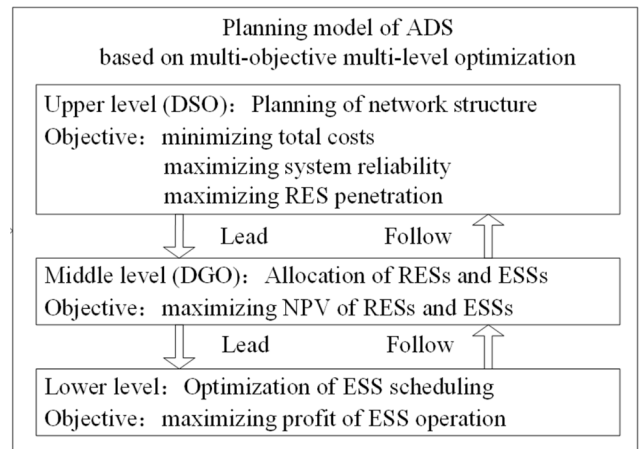


FIGURE 7. Framework of the tri-level ADNs expansion planning model (modified from [91]).

the middle level maximizes the net present value of DG owners by optimally allocating DG and ESS; and the lower level maximizes ESS operating profit via detailed scheduling, as shown in Fig 7. A similar min–max–min structure is used in [20] and [62], where the upper level selects expansion decisions, the middle level identifies worst-case realizations of uncertainties within a robust optimization framework, and the lower level minimizes generation and load-shedding costs subject to power-flow constraints and ANM schemes such as DR and network reconfiguration.

Furthermore, in [92] and [93] the three levels correspond to DSO investment, non-utility DER/DR provider decisions, and operational restoration or scheduling under contingencies; the leader's expansion plan is thus implicitly constrained by how flexibilities are procured and deployed in the lower levels. This layered structure is particularly relevant for ADNEP because it mirrors real-world governance: DSOs make long-term infrastructure decisions, DER and DR providers decide on their own portfolios and contracts, and system operation (including ANM control actions) is optimized at the lowest level given the installed assets and contractual framework.

Finally, some bilevel ADNEP models explicitly couple multistage planning with short-term corrective operation. For example, [94] integrates bi-level optimization with multistage planning by allowing long-term expansion decisions to interact directly with short-term operational responses at each planning stage. The upper level determines network reinforcements and the siting and sizing of DER and flexibility assets across multiple future periods, seeking to minimize cumulative investment and operational costs. The lower level solves an operational OPF for each stage, capturing corrective actions such as DG dispatch, ESS scheduling, and voltage control. These lower-level results determine actual operating costs, curtailment, and constraint feasibility, which feed back to the upper level and influence subsequent investment choices. By repeating this leader–follower interaction across stages, the model captures how early investments affect future operating conditions and how operational constraints drive additional reinforcements.

Similarly, [95] proposes a bilevel multistage expansion framework in which the upper level minimizes total expected cost over twenty years, and the lower level minimizes ESS investment, generation, unserved energy, DG curtailment, and DR costs through a short-term correction mechanism. In all these cases, the leader's expansion plan is “tested” by the follower under realistic operating conditions and uncertainty, making multilevel optimization a natural tool for capturing the interplay between ANM-enabled operation and long-term expansion decisions in ADNs.

4) SOLUTION METHODS

Across the reviewed ADNEP literature, the choice of solution method is shaped by the modeling objectives, the treatment of uncertainty, and the level of detail used to represent ANM and

stakeholder interactions. Authors select methods based on the trade-offs between solution quality, scalability, and modeling flexibility. Overall, mathematical programming and heuristic/metaheuristic approaches remain the most widely used for solving expansion planning problems [3], [33]. Additionally, the ADNEP models have utilized the application of various modeling tools, e.g., GAMS [25], [96] and MATLAB [19], [88], [97].

A broad range of mathematical methods and approaches have been presented to solve the ADNEP problem, including:

- Mixed integer linear programming (MILP) [26], [62], [98], [99]
- Non-linear programming (NLP) [91], [96]
- Mixed integer non-linear programming (MINLP) [20], [74], [88], [89], [93]
- Dynamic programming (DP) [63]
- Mixed integer second-order cone programming (MISOCP) [73], [100]
- Optimal power flow (OPF) [48], [75]

Meanwhile, the predominant heuristic/metaheuristic methods used across the reviewed ADNEP literature are listed below:

- Genetic algorithms (GA) [97]
- Particle swarm optimization (PSO) [91], [93]
- Fast nondominated sorting genetic algorithm [27], [28], [89]
- Immune-Genetic Algorithm (IGA) [101]
- Binary global search optimization (BGSO) [102]

Mathematical programming methods: MILP, MINLP, and MISOCP remain the most widely used, especially when precise operational detail is required (e.g., curtailment, reactive support, soft open point control, or dynamic thermal rating). Studies such as [23], [26], [98], and [103] rely on these formulations to obtain high-quality solutions. However, as models grow to include nonlinear AC power flow, energy storage dynamics, or large scenario sets, computation becomes a major challenge. This motivates the use of convex relaxations (e.g., SOCP in [100] or linearized AC power flow [104]). While these approximations reduce physical accuracy, they enable more complete integration of ANM within tractable optimization problems.

Mathematical programming models are also straightforward to implement using commercial solvers such as, e.g., CPLEX.¹ [61], [73], [98], DICOPT² [93], CONOPT [74], [75], and Gurobi³ [62], [105]. A comprehensive survey by Vahidinasab et al [38] concludes that mathematical methods generally yield more accurate results but may become inefficient when integer variables and strong nonlinearities are present.

Heuristic and metaheuristic methods are primarily chosen for their scalability and modeling flexibility [27], [28], [97].

¹IBM ILOG CPLEX Optimizer: High-performance optimization solver for linear, mixed-integer and quadratic programming.

²Discrete and Continuous Optimizer.

³GUROBI Optimization - <https://www.gurobi.com/>

These methods can explore large, nonlinear, multi-objective search spaces, but they typically rely on simplified electrical models or surrogate evaluations. As a result, heuristics tend to perform well for siting, sizing, and multi-period planning problems. Recent studies such as Pinto et al. [28] and Nasri et al. [27] show that metaheuristics can coordinate DG deployment, network reinforcement, and reliability constraints in complex planning environments, despite relying on simplified power-flow representations.

The survey by Vukobratović et al. [106] further underscores that heuristic and metaheuristic approaches are central to distribution network optimization, particularly when analytical formulations become impractical. The authors note that these methods provide pragmatic, near-optimal solutions by efficiently exploring large finite search spaces and iteratively refining candidate solutions, with established techniques such as GA, PSO, tabu search, and simulated annealing widely used in DG planning and ADN operation [106]. They also point out that limitations such as local optima or slow convergence are often mitigated through hybrid computational intelligence-based approaches integrating power flow calculations or learning models. Overall, the survey concludes that the increasing complexity of modern ADNs makes metaheuristics not only suitable but necessary for scalable and operationally meaningful optimization [106].

In more advanced planning settings, such as the multi-level active distribution system framework of Zeng et al. [97], metaheuristics provide one of the few tractable options to handle nested decision structures and uncertainties involving utilities, DG owners, demand response, and ANM schemes. Together, these findings indicate that while heuristics may rely on simplified models, they remain essential for solving large-scale planning problems where high-fidelity ANM requires complementary or hybrid optimization approaches.

The most advanced approaches in recent literature are *hybrid solution methods* emerge precisely where models become too large, nonlinear, or ANM-rich to be handled by a single “pure” solver. Several works adopt metaheuristic approaches, where a rigorous mathematical core (typically MILP or MISOCP) is combined with neighborhood search or metaheuristics: for example, [107] use a stochastic MISOCP for network reconfiguration, ESS, and DR, and wrap it in a neighborhood-based tabu search that fixes subsets of integer variables and repeatedly solves reduced MISOCP subproblems, achieving better solutions and CPU times than commercial solvers alone. Other papers implement two-level or multi-layer hybrids, where an outer metaheuristic handles long-term or discrete planning decisions and an inner exact solver optimizes short-term operations: Pamshetti et al. [94] couple PSO under the MATLAB environment for siting and sizing DERs and soft open points with a MINLP operational model solved by DICOPT in GAMS, forming a PSO+MINLP hybrid that dramatically reduces runtime compared with using DICOPT alone. Simi-

larly, [100] convert a complex ANM-rich expansion problem into a second-order cone formulation and then solve it with a two-stage hybrid algorithm, separating investment and operational subproblems to keep the global search manageable.

Survey and review papers on distribution planning emphasize that hybridizing evolutionary algorithms with classical optimization such as GA–PSO combination improves convergence robustness and enables efficient handling of mixed discrete/continuous variables in distribution planning [34]. Likewise, Vukobratović et al. [106] report that computational-intelligence methods achieve better scalability and solution quality when embedded within hybrid frameworks that incorporate mathematical programming for feasibility enforcement. This observation is further reinforced by [3] and [38], who show that hybrid metaheuristic–mathematical optimization structures increasingly dominate modern distribution planning due to their superior ability to navigate large combinatorial spaces while preserving power-flow realism. Taking together, these contributions show that hybrid methods are not an ad-hoc choice but a systematic response to the computational burden of realistic ADNEP models.

5) UNCERTAINTIES

Uncertainty in power distribution networks stems primarily from the inherent unpredictability of future system conditions, which includes variability in both supply and demand, market fluctuations, and external events [5]. In the context of ADNEP, it is crucial to incorporate one or more uncertain input parameters along with their temporal and spatial correlations to ensure realistic and robust planning outcomes under real-world conditions [30].

In the existing literature, the most addressed uncertainties are associated with short-term variations, such as hourly fluctuations in electricity demand, renewable generation (wind and solar), electricity market prices, and electric vehicle (EV) charging behavior [3], [5], [30]. These short-term uncertainties are inherently time-dependent and often spatially correlated, especially when DERs are geographically dispersed. Additionally, there is increasing recognition of the cross-correlations among different DERs such as between PV generation and load demand which can significantly influence system flexibility and operational reliability.

Conversely, long-term uncertainties, including those related to load growth, DER integration trends, policy evolution, and technological advancements, remain underexplored in most studies [3]. Addressing these long-term factors is critical for strategic grid expansion and investment planning over multi-year horizons.

Comprehensive modeling of these uncertainties can, however, lead to excessive computational complexity or even intractability [108]. Therefore, selecting an appropriate modeling technique must consider the nature of the uncertainty (e.g., probabilistic, possibilistic, bounded), its temporal char-

acteristics (short-term vs. long-term), the availability of historical data, and the computational resources available [34]. A brief discussion of these techniques is given below:

The probabilistic techniques determine the statistical properties of the uncertain parameters using the probability distribution functions (PDFs) that are based on the corresponding historical data [30]. The Monte Carlo simulation (MCS) numerical technique belongs to this classification and can handle temporal and spatial dependencies and cross-correlations. It is the most widespread technique that estimates the probability distribution functions of different uncertain parameters, e.g., the load demand [51], the availability of traditional generating units, and the intermittent renewable generators [56]. Although the probabilistic techniques adequately estimate the variability of load demand and the intermittency of non-dispatchable DGs, massive historical data is required for precise uncertainty representation.

Stochastic techniques are particularly useful when it is difficult to model uncertain parameters using probability distribution functions [30]. This technique allocates a specific probability to scenarios representing the uncertain parameter. The solution quality based on this approach depends on many scenarios; nevertheless, the computation burden would be increased. So, the challenge lies in significantly reducing the number of scenarios and selecting representative ones using algorithms such as the backward and forward scenario reduction technique, Taguchi's orthogonal array, and the clustering method [6], [38]. It should be noted that the scenario means a possible future in which uncertain parameters are specified. In this way, depending on the number of uncertain parameters, a limited number of scenarios can be defined to cover several possible events in the future. There is no specific method for defining scenarios; in the meantime, the planner's experience plays the primary role [33]. These approaches can also incorporate correlations between uncertain parameters by using joint scenario generation based on historical data.

Robust optimization is another popular technique to deal with uncertainties [19]. In contrast to the probabilistic and stochastic approaches that rely heavily on the accessibility of historical data, robust optimization represents the uncertain input parameters using the parametric bounds [30]. While seeking the best solutions that are immune to the worst-case realization of uncertainties [20]. Distributionally robust optimization goes a step further by utilizing ambiguity sets, which provide a range of possible probability distributions, thus offering a balance between the overly conservative nature of robust optimization and the precise yet often impractical requirement for exact probability distribution functions in stochastic optimization [5].

6) COORDINATED PLANNING AND OPERATION

Compared with traditional distribution networks, the ability to control DERs through ANM schemes is the most prominent feature of ADN. These ANM solutions (e.g.,

coordinated voltage control, active network reconfiguration, non-firm connection for DG and large loads, dynamic line rating, and DG curtailment), offer many potential benefits to the planning of ADNs and have become valuable alternatives to network expansion or reinforcement [13]. Therefore, the operation and planning of the active distribution network are no longer independent because the performance of ANM solutions is present in the system operation stage, which would directly impact network planning decisions. As shown in Fig. 8 to optimize the operation and planning of ADNs at once, the bi-level structure is often used to formulate the ADNs expansion planning models and achieve the coordinate optimization between planning and operation.

In the example illustrated in Fig. 8, the authors in [101] modeled the proposed ADNs planning approach as a bi-level optimization problem. The first level (corresponding to the decision-making in the planning phase) comprises network reinforcement/expansion and planning of DGs (wind units) and ESSs. The decision variables of this level include the optimal timing, location, and type/number of the system lines and substation/DG units to be upgraded or installed.

The second level is defined as a two-stage optimization problem where the operational cost and the energy not supplied (ENS) cost are calculated. In the first stage, distribution system companies (DISCOs) determined the optimal operational behavior using combined data (including network data received from the upper level, forecasted generation-demand data, and day-ahead market prices). The optimal operational behavior includes the optimal power generation of the DGs, and the amount of power purchased from the transmission network for each network state. These are achieved through the execution of the proposed MILP-based OPF that considers environmental and economic objectives simultaneously.

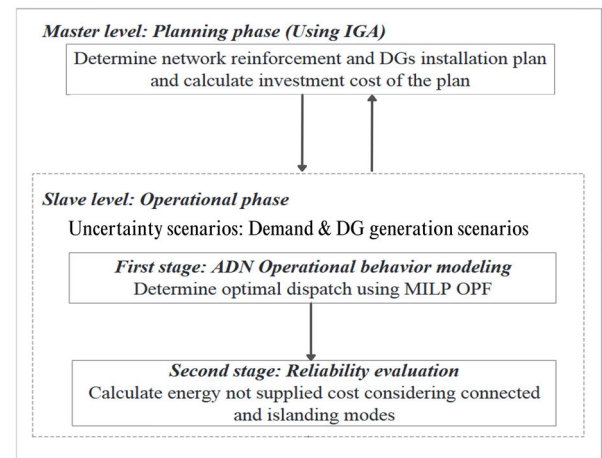


FIGURE 8. Example of a framework of the bi-level ADNs expansion planning model(modified from [101]).

Meanwhile, the feasibility of the first-stage decisions is checked in the second stage, where the technically infeasible

solutions are identified and discarded. In the second stage of the second level, the generated operational behavior is utilized to analyze the candidate planning proposal's reliability quantitatively. At this stage, the ENS cost is assessed considering both the connected and island mode operations. DG units can supply the loads during scheduled or unscheduled outage events within islands. The estimated operational cost, ENS cost along with the economic risk of the planning scheme will be feedback to revise previous planning design in the first level. With ongoing iterations, this procedure would finally reach the optimal plan.

D. LITERATURE REVIEW ON ACTIVE DISTRIBUTION NETWORK EXPANSION PLANNING

Numerous methods addressing various forms of ADNEP have been proposed over the past decade, as summarized in Tables 1 and 2 and illustrated in Fig. 9. Table 1 provides a categorical overview of selected representative studies, organized according to characteristics such as planning time horizon, DER integration, uncertain input parameters, solution methodologies, and the modeling platforms or solvers employed. Table 2, on the other hand, classifies the reviewed literature based on optimization objectives, reliability indices, and key contributions. Although ANM technologies and services are analyzed separately in Section III, the references that incorporate ANM solutions are emphasized in **bold** in both tables.

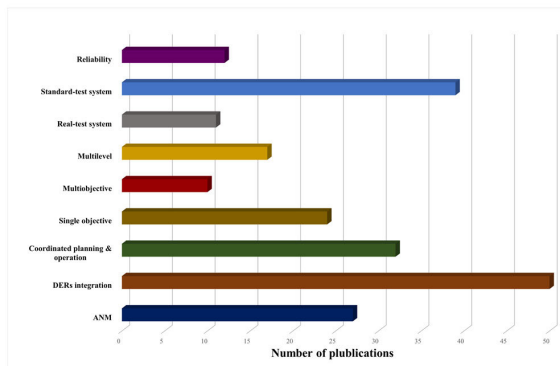


FIGURE 9. Distribution of publications addressing key elements in ADNEP framework.

As detailed in Table 1, the second column categorizes the planning time scale adopted in the reviewed studies, distinguishing between single-stage and multi-stage formulations. Despite the added complexity introduced by interdependence between planning stages, multi-stage formulations are predominant in the reviewed studies. Their widespread adoption reflects a growing consensus on their ability to produce more practical and forward-looking solutions, especially in the dynamic and uncertain environment characteristic of modern ADNs.

The third column of Table 1 addresses the integration of distributed energy resources in the expansion planning models. Only a minority of the reviewed studies have mod-

eled the combined integration of all three DER components within a single framework. Most studies included only one or two DER elements, with the DG–ESS pairing being the most adopted. This limited scope suggests that the synergistic benefits of full DER integration are not yet fully leveraged in current expansion planning models and remain a promising direction for further investigation.

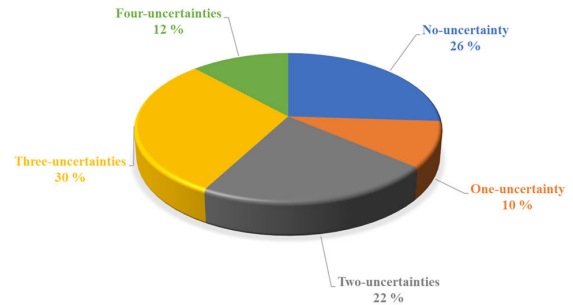


FIGURE 10. Summary of the reviewed uncertainty parameters.

Among the studies reviewed in this work, seventeen articles incorporated DR mechanisms into expansion planning models, as illustrated in Table 1, third column. Demand response can be implemented through various mechanisms, such as load curtailment, load shifting at low-cost hours, and a bilateral contract between DSO and customer (e.g., a flexible connection contract). Another way is to participate in the local flexibility market, where DR is defined as a service traded on the market.

For instance, the work presented by [18] the DR capacities from the responsive customers are purchased by the DR provider (“aggregator”) as the market player and negotiated with the DSO to sign a DR contract to sell these potentials to maximize the DR provider profit. Meanwhile, in [74] the authors incorporated the control capabilities of active management of DG units and the control capabilities of load demand via demand response into the solution of the planning problem. It is assumed that assigned consumers can follow demand response schemes, which means they can adjust their hourly load level if needed.

In the reviewed literature, most studies assume that the location and capacity of renewable and non-renewable DGs, ESSs, and capacitor banks are known, as summarized in Table 1, column four. This assumption is facilitated by tools such as the Global Wind Atlas⁴ and Photovoltaic Geographical Information System⁵ (PVGIS) which already provide high-resolution data identifying suitable locations for wind and solar energy based on resource availability. When this data is combined with land-use restrictions, proximity to grid infrastructure, and transportation access, a highly informed estimation of potential renewables DG locations becomes feasible. However, it is important to note that final investment decisions are typically made by DG owners, prosumers,

⁴<https://globalwindatlas.info/en/>

⁵https://joint-research-centre.ec.europa.eu/photovoltaic-geographical-information-system-pvgis_en

and other market stakeholders, not by distribution system operators.

Furthermore, column 5 highlights the uncertain parameters considered in the ADNEP problem. As illustrated in Fig. 10, more than 60% of the articles reviewed addressed the combination of several uncertain issues, such as load demand, DG generation, and electricity prices. For instance, in the work presented by [23] they addressed the uncertainty associated with power demand, renewable generation, and electricity prices in the stochastic model for active distribution network planning.

Column 6 of Table 1 illustrates modeling techniques to deal with uncertainties. The stochastic approach is widely used in the references. The authors in [112] proposed a stochastic framework to address the joint expansion planning of a restructured active distribution network while considering the profitability of the attained expansion plan for distributed generator owners (DGOs). Furthermore, DSO takes uncertainties associated with DGOs' investment budget into account.

Meanwhile, the paper presented in [114] models uncertainties in photovoltaic (PV) generation and load using a data-driven stochastic modeling technique based on Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP) combined with the K-means clustering method. The WGAN-GP is used to generate realistic uncertainty scenarios of PV output and load profiles that capture the stochastic and correlated nature of renewable generation and demand. Additionally, K-means clustering is then applied to reduce the number of generated scenarios into a representative set of scenarios, balancing model accuracy and computational efficiency.

From column 7, it can be appreciated that most of the expansion planning models have followed the basic structure of a single objective function. Contrary to the fact that real-world planning problems are multi-objective in nature [49], [115]. Nevertheless, a raising number of articles implemented the multi-objective, bilevel and tri-level optimization perspective.

For instance, the authors in [103] propose a resilient expansion planning framework that compares dynamic line rating (DLR) with conventional line reinforcement under low-probability, high-impact (LPHI) outages. The model incorporates line hardening, DLR device installation, and the optimal formation of radial reconfigurable microgrids, and formulates a multi-objective optimization that minimizes total cost such as construction, operation, and CO₂ emissions alongside load shedding. Implemented as a convex

mixed-integer quadratic-constrained program in GAMS and tested on the IEEE 24-bus system, the approach generates Pareto-optimal solutions, with the final plan selected via a fuzzy-satisfying method. Results show that the proposed strategy achieves a resiliency index above 92% under LPHI events, offering practitioners a robust tool for evaluating DLR versus reinforcement and providing policymakers with quantitative insights into emissions–reliability trade-offs.

Column 8 shows that several articles use MINLP as the solution method. This technique combines linear programming, mixed integer programming, and nonlinear programming [49]. This approach proves better computation efficiency and accuracy and provides correct, reliable, and efficient solutions for multi-objective formulation although it is difficult to implement in nonlinear objective functions and with much iteration. Furthermore, due to the complexity of the ADNEP problem, most of the analyzed articles use more than one solution method in combination with optimization solvers and modeling tools, as shown in column 9.

According to the summary presented in Table 2, more than 50% of the reviewed studies adopt a coordinated approach to expansion and operation planning. This trend reflects a growing recognition that decoupling these two stages can lead to suboptimal and inflexible network development strategies. Furthermore, the minimization of investment and operational costs emerges as the most addressed objective function across the literature, highlighting the economic efficiency focus in ADNEP methodologies.

Notably, [18] introduces a stochastic system-of-systems (SoS) architecture for adaptive expansion planning, which represents a significant departure from traditional monolithic approaches. In this framework, SoS is defined as a system containing a set of autonomous and heterogeneous systems with discrete and in some cases conflicting objective functions, who have interoperability and are aggregated for a common goal. The model acknowledges that while each system seeks to maximize its individual profit, inter-system cooperation and shared decision variables are necessary to ensure global feasibility and system-level optimization.

Specifically, [18] models three distinct decision-making entities: the distribution company, a private investor, and a demand response DR provider. Within this framework, the private investor is responsible for determining an optimal plan for DER deployment, while the DR provider focuses on formulating profitable DR contracts. Concurrently, the distribution company aims to ensure a cost-effective and reliable grid upgrade. The objective functions of each entity, which are designed to capture their individual interests while enabling system-level coordination, are summarized in Table 2, column 3.

1) RELIABILITY ANALYSIS IN ADN EXPANSION PLANNING PROBLEM

Societies are increasingly dependent on secure electricity supply due to digitalization and electrification. Considering this and the fact that the provision of service reliability typi-

⁶Departure and arrival times, initial and target energy of AEVs.

⁷It represents the cost of unserved energy during power interruptions [116].

⁸Loss of load cost represents the cost of not supplying the energy to a specific customer [28].

⁹The proposed model is formulated as a tri-level optimization problem, and it is solved using a two-stage robust optimization approach using MILP.

TABLE 1. Summary of the solution methods and others features in the reviewed ADNEP models.

Ref.	Time scale	DERs	Size /Location	Uncertain parameters	Modeling techniques	Optimization	Solution methods	Modeling tool/Solvers
[98]	Multi-stage	DG	DG	Wind generation and load demand	Probabilistic approach	Single objective	OPF-NLP	GAMS/ CONOPT
[22]	Single stage	DG & DR	DG	Load demand, wind & PV generation	Stochastic	Multiobjective	MINLP	GAMS/ DICOPT
[57]	Single stage	DG & DR	DG	PV generation and load demand	Monte Carlo	Multiobjective with weights	-	GAMS/ IPOPT
[104]	Multi-stage	DG	DG	Load demand	Monte Carlo	Single objective	BGSO	-
[99]	Multi-stage	DG & DR	DG	Load demand, wind & PV generation	Probabilistic approach	Tri-level	OPF-MINLP-GA	MATLAB
[25]	Multi-stage	DG & DR	DG & CB	-	-	Single objective	MINLP	GAMS/ SNOPT
[49]	Multi-stage	DG	DG	-	-	Single objective	OPF	-
[111]	Single stage	DG	DG	Wind generation and load demand	k-means clustering	Single objective	MINLP-GA	-
[60]	Multi-stage	DG	DG	-	-	Single objective	OPF-MINLP	FICO Xpress
[77]	Multi-stage	DG	DG	Load demand, wind & PV generation	Monte Carlo	Single objective	OPF	GAMS/ CONOPT
[21]	Multi-stage	DR & ESS	ESS	-	-	Single objective	MILP	MATLAB/ Gurobi
[59]	Multi-stage	DG	DG	-	-	Multiobjective with weights	MINLP-GA	MATLAB-GAMS/ CONOPT3
[93]	Multi-stage	DG & ESS	DG & ESS	PV generation and load demand	k-means clustering	Tri-level	NLP-PSO-NSGA II	-
[112]	Single stage	DG	DG	Load demand, wind & PV generation	-	Multiobjective	Harmony Search	MATLAB
[103]	Multi-stage	DG & ESS	DG & ESS	Wind generation	Stochastic	Bilevel	MILP-OPF-IGA	-
[107]	Multi-stage	DG	DG	-	-	Single objective	MILP	Gurobi
[113]	Multi-stage	DG & DR	DG	-	Stochastic	Multiobjective	MINLP	MATLAB/ CPLEX
[52]	Multi-stage	DG & DR	Switches	Load & EV demand, wind, and PV generation	Monte Carlo	Single objective	Linear programming	GLPK
[18]	Multi-stage	DG & DR	DG	Wind generation, load demand and DR potential	Probabilistic approach and Stochastic	Multiobjective	MINLP- PSO	-
[26]	Multi-stage	DG & ESS	ESS	Load demand, wind & PV generation	Stochastic	Single objective	MILP- OPF	MATLAB/ CPLEX
[100]	Multi-stage	DG	DG	Wind generation and load demand	Stochastic	Single objective	MILP	GAMS/ CPLEX
[91]	Multi-stage	DG & ESS	DG & ESS	Load & EV demand, wind, and PV generation	Probabilistic approach	Bilevel (Multiobjective)	MINLP-Fuzzy-NSGA II	MATLAB
[74]	Multi-stage	DG & ESS	-	Wind & PV generation	Robust optimization	Bilevel	MISOCP	MATLAB/ CPLEX
[28]	Multi-stage	DG	DG, CB & Switches	-	-	Multiobjective	MINLP-OPF-NSGA II	MATLAB
[19]	Multi-stage	DG & ESS	DG, CB & ESS	Wind & PV generation	Stochastic	Single objective	MISOCP- CCG	MATLAB/ CPLEX
[94]	Multi-stage	DG, DR & ESS	-	Wind & PV generation, load demand, and electricity price	Robust optimization	Bilevel	Tri-layer decomposition method	-
[101]	Multi-stage	DG & ESS	DG & ESS	Wind generation and load demand	Stochastic	Single objective	MILP	CPLEX
[90]	Multi-stage	DG & ESS	DG & ESS	PV generation and load demand	Stochastic	Bilevel	MINLP	GAMS/ CPLEX
[58]	Single stage	DG & ESS	SEVCS, DG & ESS	Shared EV's start charging time and charging duration.	Stochastic	Single objective	NAA	-
[102]	Multi-stage	DG & ESS	DG & ESS	Load demand, wind & PV generation	Stochastic	Single objective	MISOCP	CPLEX
[24]	Single stage	DG & ESS	DG & ESS	-	-	Single objective	Linear programming	-
[20]	Multi-stage	DG, DR & ESS	DG, CB & ESS	PV generation and load demand	Robust optimization	Tri-level	MILP-CCG-ARO	Gurobi
[75]	Multi-stage	DG & DR	DG	Load demand, wind & PV generation	Stochastic	Single objective	NLP-GA	CONOPT
[95]	Multi-stage	DG, DR & ESS	DG & ESS	Electricity price, NDERs power generation pattern, Gas engine electricity generation, NDERs time and installation location	Stochastic	Tri-level	MINLP-PSO	DICOPT
[62]	Multi-stage	DG & ESS	DG, CB & ESS	Wind & PV generation, load demand, and electricity price	Stochastic	Single objective	MILP	CPLEX
[27]	Multi-stage	DG	DG	-	-	Multiobjective	MINLP-NSGA II	MATLAB
[63]	Multi-stage	DG & ESS	DG, CB & ESS	Load demand, wind & PV generation	Robust optimization	Tri-level	MILP - OPF - ARO	GUROBI – IPOPT
[64]	Single stage	DG & ESS	DG, CB & ESS	-	-	Single objective	DP - Deep neural networks	Gurobi
[106]	Multi-stage	DG, DR & ESS	DG & ESS	-	-	Single objective	MILP	GAMS/ CPLEX/ CONOPT
[96]	Multi-stage	DG & ESS	DG, SOP & ESS	Load demand, wind & PV generation	Stochastic	Bilevel	MINLP-PSO	GAMS/ MATLAB
[92]	Multi-stage	DG & ESS	-	Load demand, wind & PV generation	Probabilistic approach	Bilevel	MILP	GAMS/ CPLEX
[114]	Multi-stage	DG & ESS	DG & ESS	Agents' investment budget	Stochastic	Single objective	MILP	GAMS/ CPLEX
[23]	Single stage	DG	DG, CB & SVC	Wind & PV generation, load demand, and electricity price	Stochastic	Single objective	MISOCP	Gurobi

TABLE 1. (Continued.) Summary of the solution methods and others features in the reviewed ADNEP models.

[78]	Single stage	DG	DG	PV generation, load demand, and weather factors	Stochastic	Single objective	MBD	Gurobi/ MATLAB
[115]	Single stage	DG, DR & ESS	DG, CB, ESS & SVC	Behavioral characteristics of AEVs ^o	Stochastic	Bilevel	MILP-MISOCP	Gurobi/ MATLAB
[109]	Single stage	DG, DR & ESS	DG, CB, ESS	PV generation, load demand, and energy costs	Stochastic	Single objective	MISOCP-NMA	Cplex/ Gurobi
[105]	Single stage	DG	DG	-	-	Multiobjective	MIQCP-Fuzzy	Gurobi/ GAMS
[116]	Single stage	DG & ESS	DG, ESS & SVC	PV generation and load demand	Stochastic	Bilevel (Multiobjective)	Adaptive PSO - MISOCP	Gurobi/ MATLAB
[79]	Single stage	DR	EVCS	-	-	Bilevel	PSO-SOC	-
[97]	Multi-stage	DG, DR & ESS	DG & ESS	Load demand, wind & PV generation	Probabilistic approach	Bilevel	Prim-Dijkstra hybrid algorithm	Gurobi/ MATLAB

Notes - CCG: column & constraint generation algorithm; SEVCS: shared EV charging stations; NAA: natural aggregation algorithm; ARO: adaptative robust optimization algorithm; NDERs: non-utility DERs; SVC: static var compensators; CB: capacitor banks; SOP: soft open points; MBD: modified Benders decomposition; AEVs: autopilot electric vehicles; MIQCP: convex mixed-integer quadratic-constrained programming; SOC - second-order cone relaxation; EVCS: EV charging stations; SOCP: second-order cone programming; NMA: neighborhood-based metaheuristic approach

cally accounts for almost half of the total distribution system costs [117], several indices are used to quantify the reliability of distribution networks (e.g., SAIDI, SAIFI, and EENS). The evolution of distribution networks towards active ones has created a new set of opportunities for further improvement of the reliability of supply. For example, during a fault in the ADN an aggregation of both renewable and non-renewable DGs can provide sufficient power to supply those interrupted areas that cannot be otherwise supplied by primary substations. Energy storage technologies can be used to mitigate the fluctuations of renewable generation and extend their contribution to supply restoration. Automation of the protection devices can be used to reduce the time response necessary for the network reconfigurations in presence of fault conditions. Also, the application of DR mechanisms can help decrease the peak demand selectively and preserve the security of supply under emergency conditions [118].

Therefore, considering reliability in expansion planning models of the ADN becomes of crucial importance for the planners. A thorough literature survey of the state of the art regarding models and methods applied to the solution of the distribution system expansion planning problem considering reliability is given by Aschidamini et al. [32]. In their conclusion, the authors in [32] state that system reliability can be significantly improved when DER are included in the models. Furthermore, including switching devices capable of locating faults can substantially improve reliability indices. However, as more reliability-enhancing strategies are envisioned, the complexity of the optimization model inevitably increases.

Several publications have addressed reliability in expansion planning models, as shown in Table 2. The articles presented by [18], [89], and [91] assessed the reliability through the minimization of EENS as the second objective function of the model. In [18] the reconfiguration of the ADN as part of the planning alternative to minimize the EENS is implemented. Publication [91] estimates EENS by sequential Monte Carlo Simulation, in the process of calculation, the features of uncertainty and contingency of renewable energy sources and ESSs are taken into account adequately. Meanwhile, the paper presented by [89] the

two-state Markov process model serves to simulate sequence transitions between operative states and failed states of main components, including wind generators, photovoltaic, energy storage units and feeders. In conclusion it is stated that based on the proper allocation and optimal operation, energy storage units are an effective and feasible way to improve economy efficiency and reliability.

On the other hand, in [28] the cost related to the reliability is calculated as part of the first objective function and SAIDI index is defined as the second problem objective, in order to avoid only monetary factors defining expansion planning. The Monte-Carlo simulation method was chosen to estimate the SAIDI index, and it is supposed that switches perform self-healing properly.

In [62] the proposed model uses a three-level decomposition framework and aims for a global solution for the expansion problem considering that the DSO owns the DERs' expansion decision, which is subject to uncertainties in generation capacity and demand. The adaptive robust optimization algorithm developed seeks the best investment proposal that would meet the EENS target and minimize the total cost of the problem (investment, generation, and load shedding costs) for the worst-case realization of the uncertainties.

Publications [26], [93], considered the reliability through minimization of EENS costs as part of the objective function. The work presented in [26] besides EENS costs, SAIDI and SAIFI indices are considered in the model as constraints and the network planner can define desired target values for those two indexes to impose an improvement.

A realistic 180-bus distribution network, which is a part of the distribution network located in Leiria district in Portugal was used to test the proposed stochastic model in [26] and compared with a traditional method (deterministic) to demonstrate the advantage of the former. It was verified that the proposed stochastic method presents a reduction around 35.54% of the total cost compared with the deterministic approach. On the other hand, the authors in [21] proposed a model where three indices on the economics of construction, electricity trading, and reliability are included in the objective function to optimize the long-term investment cost

TABLE 2. Objective functions and main contributions of the reviewed ADN expansion planning models.

Ref.	Planning type	Objective	Reliability	Contributions
[98]	Planning & operation	Maximize the total DG capacity	-	Provides detailed analysis and results on how multi-DG interactions could impact the amount of connected DG capacity as well as the utilization of ANM schemes. Proposes a new OPF-based planning approach which considers the operational status of DG units at the planning stage and evaluates the DG capacity considering various multi-DG configurations.
[22]	Operation	<i>First objective:</i> Minimize operation cost <i>Second objective:</i> Minimize emission	-	A new conceptual design of an intelligent DR management system integrated with an advanced metering infrastructure system to support DERs management in the environment of future distribution systems. A new method for energy and reserve scheduling in distribution networks is proposed as one of DR management system applications.
[57]	Expansion & operation	Minimize the total operational cost and the total energy losses	-	A Monte Carlo simulation-based multi-objective optimization approach which considers DR and ANM schemes at the planning stage
[104]	Expansion	Minimize investment, operation and losses costs	-	Multistage planning procedure is proposed to consider the pseudo-dynamic behavior of planning.
[99]	Planning & operation	<i>Upper Level:</i> minimize investment and operation costs <i>Middle level:</i> maximize net profit of DGO <i>Lower level:</i> maximize profit of ESSs operation	-	A novel multi-level optimization model is developed to formulate the problem, wherein the operational characteristics are properly coordinated in the planning setting.
[25]	Expansion	Minimize annualized capital and operation costs	-	A comprehensive multi-year distribution system planning framework that considers DR as a plan option to mitigate the impact of plug-in EVs charging loads. A novel set of plug-in EVs operational constraints are included in the distribution planning model.
[49]	Expansion & operation	Minimize the losses, active power curtailment and reactive power control costs.	-	A methodology to co-optimize the planning and operating phases of modern distribution networks with active elements.
[111]	Expansion	Minimize NPV	-	A hybrid GA – NLP approach for the planning of active distribution networks.
[60]	Expansion & operation	Minimize capital and operational costs	-	An AC-OPF algorithm to optimize the network reinforcement plan while maximizing the use of existing assets and actively managing the real-time operation of the network.
[77]	Operation	Maximize social welfare	-	A novel multi-configuration multi-period market-based optimal power flow.
[21]	Expansion & operation	Minimize the total investment and operation cost	Value of lost load ⁷	A multi-stage ADN planning model that applies the MILP approach for integrating ESSs. The benefits of centralized ESSs and distributed ESSs in the ADN planning are discussed and evaluated for improving the reliability of the grid.
[59]	Expansion & operation	Minimize investment cost, feeders load balance and DG curtailment	-	A multistage coordinated planning framework of active distribution networks in the presence of high renewable DG integration. Incorporation and analysis of various ANM schemes on the distribution network planning solution.
[93]	Expansion & operation	<i>Upper Level</i> (three objectives): minimize investment, operation and maintenance costs; minimize EENS; maximize RESs penetration <i>Middle level:</i> maximize NPV of DGO <i>Lower level:</i> maximize profit of ESSs operation	EENS	The planning model considers the optimization of ESS adequately. It achieves cooperative optimization between planning and operation in different time scales. The proposed model provides a set of Pareto alternative planning schemes from different perspectives including economy, reliability, and utilization of RESs.
[112]	Operation	<i>First objective:</i> Minimize total annual energy loss <i>Second objective:</i> Minimize voltage deviation <i>Third objective:</i> Minimize cost of DG integration	-	New hybrid Grid-based Multi-objective Harmony Search algorithm is proposed by incorporating the grid setting for selection of best objectives value. Pareto based multi-objective DG planning model is proposed for active distribution network operation with three conflicting objectives.
[103]	Expansion & operation	<i>Upper Level:</i> minimize NPV of the total investment costs. <i>Lower level</i> (two stage): optimal operational cost along with the ENS cost <i>First stage:</i> minimize operational cost <i>Second stage:</i> reliability evaluation model	ENS	ESSs and renewables DGs are concurrently incorporated into the ADN expansion planning problem in which the contribution of these resources in reliability enhancement of the network is considered by modeling island mode operation. A bi-level optimization model is developed; the proposed model combines the system planning and operational decision-making process under the same framework. The economic risk associated with the stochastic input parameters is considered as a planning objective to assure that the system could have desired economic risk level after the integration of the uncertain resources.
[107]	Expansion & operation	Minimize the total investment and operation cost	-	Propose a multi-stage expansion planning method considering DN reconfiguration and even operation of microgrids using DGs for ADN, in which the infrastructure investment options, and operation strategies of each planning stage can be co-optimized.
[113]	Expansion	<i>First objective:</i> Minimize present total cost <i>Second objective:</i> Minimize the minimum spanning tree	-	A novel multi-objective model with multiple ANMs schemes for active distribution network planning based on stochastic programming and uncertain random network theory is proposed.
[52]	Expansion	Minimize capital expenditure of the network	-	To integrate time series curves instead of considering only the worst case (peak consumption), which could be different in the presence of DGs and EVs. To model and integrate advanced functions such as reconfiguration and DR in the long-term planning study
[18]	Expansion & operation	<i>Distribution company</i> <i>First objective:</i> Minimize planning costs <i>Second objective:</i> Minimize EENS <i>Private investor objective:</i> Maximize profit (selling the power of DERs) <i>DR provider objective:</i> Maximize profit (selling the DR potentials)	EENS	Proposing the system of systems architecture to model the independent behavior of the distribution company, private investor, and DR provider. The incorporation of the distribution system reconfiguration with the expansion planning, while the potentials of DR and DERs are considered.
[26]	Expansion	Minimize total expected planning cost	EENS; SAIDI; SAIFI	Considering the seasonal (spring, summer, fall and winter) and daily periods (night, morning, peak and afternoon) impact effect in a long-term distribution network planning in a smart grid context with high-RES penetration
[100]	Expansion & operation	Minimize NPV	-	A joint multistage distribution network planning in which the active management of resources is addressed.
[91]	Planning & operation	<i>Upper Level</i> (three objectives): Minimize total costs; minimize EENS; maximize renewable DG penetration <i>Lower level</i> (two objectives): Maximize net profit; maximize reserve capability of ESSs	EENS	To optimize planning issues and operation issues cooperatively, the operation strategy of ESSs is formulated as a fuzzy multi-objective optimization model and incorporated into the bi-level planning model, which enables an adequate consideration of ESSs operation issues and further improves the quality of planning solutions.
[74]	Expansion & operation	<i>Upper level:</i> Minimize total cost <i>Lower level:</i> Max-min operation cost	ENS	A bi-level robust planning model for active management elements including on-load tap changers, electrical storage systems, capacitor banks, and static VAR compensations.
[28]	Expansion & operation	<i>First objective:</i> Minimize expansion, operation, and reliability costs. <i>Second objective:</i> Minimize SAIDI index	EENS; SAIDI; Loss of load cost ⁸	Multi-objective and multi-period distribution expansion planning modelling, considering simultaneously, DG, capacitors, switches, branches, network reliability analysis, and self-healing.
[19]	Planning & operation	Minimize the total investment and operation costs.	-	A joint planning model considering DGs with reactive power output, distributed ESSs and different types of controllable loads is proposed.
[94]	Expansion & operation	<i>First layer:</i> Minimize the total investment and equivalent operating costs. <i>Second layer:</i> Minimize wind and PV generation curtailments <i>Third layer:</i> Minimize the variance of nodal net load	-	A generalized double-nested game model for formulating a cost-effective robust expansion planning scheme. A closed-loop scheduling mechanism combined with dynamic pricing based on a multi-sided trading pattern. A tri-layer decomposition method for solving the presented nested game model which cannot be solved directly by off-the-shelf tools.

TABLE 2. (Continued.) Objective functions and main contributions of the reviewed ADN expansion planning models.

[101]	Expansion & operation	Minimize NPV	-	Multistage planning model in which the active management of network and DERs, including ESSs, is addressed.
[90]	Planning & operation	<i>Upper level:</i> Maximize the total profit of the ICES <i>Lower level:</i> Maximize each prosumer utility	-	A bilevel operation-planning model to handle ESSs and PV planning while considering operational conditions. Applying an excess-demand penalty to supervise the contract between the ICES and the WM.
[58]	Expansion	Minimize the total annual investment cost, energy loss, and shared EV waiting cost	-	Consideration of the EV sharing systems in the long-term joint ADN planning problem. A scenario analysis method that fully considers the stochastic feature of shared EV charging loads.
[102]	Expansion & operation	Minimize the total investment and operation costs	-	A comprehensive expansion planning model of ADNs with consideration of full active network management schemes. Demand management accounting for the optimal load-shedding direction.
[24]	Expansion	Maximize the overall profit to the DSO	-	A planning model for active distribution networks is proposed considering distributed energy resources.
[20]	Expansion & operation	<i>First stage:</i> Minimize the investment costs. <i>Second stage:</i> Max-min of the total generation and load shedding costs	-	A novel tri-level ⁹ min-max-min robust optimization model for the coordinated operation and expansion planning problem of ADNs with multiple microgrids including contingency. The inclusion of demand response in a robust model, where the load is represented by three different portions (non-controllable, controllable, and deferrable). The representation of the daily 24-hour operation inside the expansion planning using a robust model and contingencies.
[75]	Expansion	Minimizes NPV of the total investment and operational cost	-	A multistage planning framework of ADNs with high penetration of RESs that considers multiple planning alternatives and the uncertainties of load demand and renewable generation.
[95]	Expansion	<i>First level:</i> Minimize total investment and aggregated operation costs <i>Second level:</i> Maximize the benefit of NDERs <i>Third level:</i> minimizes the expected value of UDERs/DRPs costs	ENS	A multi-stage optimization framework is proposed for the resilient electric distribution system expansion planning problem. The problem evaluates the market power of NDERs in the restoration plans in contingency conditions.
[62]	Expansion	Minimize investment and operation costs	-	A novel two-stage stochastic programming model is presented to address the medium-term reinforcement planning of active distribution networks considering multiple investment alternatives, network reconfiguration, and CO ₂ emissions limit.
[27]	Expansion	<i>First objective:</i> Investment and operation costs <i>Second objective:</i> Resilience index	-	A multi-stage and dynamic framework are presented to resiliently expand and reinforce a distribution network against the hurricane. An innovative model is proposed by considering a novel vulnerability index based on the line fragility curve, line importance, and line deference angle from the hurricane path.
[63]	Expansion	<i>First level:</i> minimize total cost (investment, generation, and load shedding) <i>Second level:</i> minimize the generation and load shedding costs. <i>Third level:</i> MCS is used to accurately calculate the EENS index	EENS	A robust expansion planning model for active distribution networks representing uncertainty and reliability. A novel sensitivity matrix to incorporate reliability indexes into the expansion planning problem. Inclusion of switches and self-healing analysis inside a robust expansion planning model.
[64]	Operation	Minimize the annual operational costs	-	Instead of solving the model-based optimization problem with fixed solution spaces, this paper models the dynamic update of the solution spaces of the AND planning scheme as a Markov decision-making process.
[106]	Expansion & operation	Minimize overall network costs, encompassing both investment and operational expenses	-	Development of a two-stage deep learning-based methodology for optimal sizing and siting planning of wind turbines, PV systems, and battery ESS in the distribution network. Analysis of the impact of ANM solutions, and system flexibility achieved through tie-line investments on the sizing and siting decisions for RESs in active distribution networks. Linearization of the mathematical model is used in the optimization method to enhance its scalability.
[96]	Planning & operation	<i>Upper level:</i> Minimizes the NPV of wind, solar plant, battery ESS and SOP investment costs <i>Lower level:</i> Minimizes the NPV of total projected variable costs for all credible system operating states	ENS	For an integrated planning model of PVs, wind generators, battery ESS, and SOP in an active distribution network, a two-stage coordinated optimization methodology was proposed. In ADN, the combined influence of network reconfiguration and conservation voltage reduction approaches on PVs, wind generators, battery ESS, and SOP planning was investigated.
[92]	Expansion & operation	<i>Upper level:</i> Minimizes the present value of total cost imposed on all stakeholders <i>Lower level:</i> Minimizes the total generation costs	-	The ADN strategic visioning problem is formulated as an expansion planning model from the perspective of electric distribution stakeholders' organization, as a non-monopolistic non-profit operator, using a multistage bi-level framework. The multistage expansion problem is solved considering the local energy market at the ADN level in the presence of various active customers with conflicting interests. The active customers and load uncertainties are considered in the market clearing as well.
[114]	Expansion & operation	Minimizes the NPV of the total costs associated with investment, maintenance, and operation of the network	-	A multistage expansion planning of a restructured ADN is addressed in which the techno-economic constraints of all participating agents are considered while the uncertainty associated with the agents' investment budget is modeled via a stochastic optimization framework.
[23]	Planning & operation	Minimizes total investment and network operating costs	-	A novel MISOPC model is proposed for ADN planning that integrates various ANM schemes.
[78]	Expansion & operation	Minimize the total investment and operation costs	-	We propose a stochastic DNEP model of ADNs incorporating the DTR of cables and transformers, which integrates the thermal models of cables and transformers to formulate the operational constraints with the installation of DTR equipment. To effectively solve the proposed nonlinear DNEP model, we derive a linear reformulation of the original model and propose a MBD algorithm capable of accelerating the convergence procedure.
[115]	Expansion & operation	<i>Subproblem I:</i> Minimizes AEV charging costs. <i>Subproblem II:</i> Minimizes annualized investment cost, operational and maintenance expenses, and power system network loss.	-	A novel collaborative planning model for joint deployment of the ADN and V2GCS. This model takes into full consideration V2G functionality of EVs and efficiently utilizes unused capacities of V2G inverters to provide reactive power support for the grid. A sequential decomposition method is proposed, transforming the holistic problem into two sub-problems, based on the weak coupling property of the physical problem.
[109]	Operation	Minimizes the active power losses, CO ₂ emissions, rewards for the demand response program, and energy not supplied to loads	ENS	Introducing a pioneering mathuristic algorithm that seamlessly integrates mathematical programming and heuristic techniques. This combination enables the effective operational optimization of complex distribution systems, considering diverse operational aspects. Emphasizing specific attention to challenges inherent in modern distribution systems, such as network reconfiguration, dispatch of fossil fuel-based and renewable DG, DR programs, and ESS.
[105]	Expansion & operation	<i>First objective:</i> Minimizes distribution system expansion planning cost <i>Second objective:</i> Minimizes load shedding	-	Integration of DLR equations into sustainable distribution network expansion planning at the presence of renewable and non-renewable DG units. Simultaneous DLR placement and feeders' reinforcement decisions by presenting new convex formulations. Proposing a reconfigurable radial micro-grid-based operation for the system under emergency conditions.
[116]	Expansion & operation	<i>Upper level:</i> Minimize the total investment cost of flexible resources <i>Lower level (three objectives):</i> Minimizes annual operational costs, network loss, and voltage deviation	-	A novel bi-level collaborative planning model considering multiple objectives with different attributes such as investment cost, voltage deviation and network loss is proposed. This paper cooperatively configures EV charging stations with orderly charging guidance, ESSs and active management resources (SOPs, SVCs, and OLTC), aiming to solve the problem of high-penetration PV integration while minimizing cost and keeping flexibility for distribution network operators.
[79]	Expansion & operation	<i>Upper level:</i> Minimize the total investment and operation costs <i>Lower level:</i> Minimizes operation cost	-	The proposed bilevel planning model considers active response EVs while incorporating the dynamic thermal rating of cables and transformers, thereby reducing investment costs and accommodating the growing load demand. An iterative solution framework was established, where Prim algorithm is embedded into the PSO algorithm in the upper level and second-order cone relaxation and linearization of the variable product are carried out for the non-linear constraints of cables and transformers in the lower level.

TABLE 2. (Continued.) Objective functions and main contributions of the reviewed ADN expansion planning models.

[97]	Expansion & operation	Upper level: Minimize the total expected cost Lower level: Minimize ESS investment cost, maintenance, generation, and unserved energy costs, DG curtailment penalty and demand response cost	ENS	A bilevel multistage expansion planning for distribution networks model incorporating a short-term correction mechanism is developed to coordinate multiple uncertainties and enhance the adaptability of long-term planning to operational needs. An adaptive phase partitioning strategy that considers source-load dynamic growth and stochastic distribution is proposed. A key node assessment method is proposed to optimize the candidate pool for ESS siting, prioritizing nodes that require targeted investment management.
------	-----------------------	---	-----	---

Notes: NPV - Net present value; EENS - Expected energy not served; SAIDI - System average interruption duration index; SAIFI - System average interruption frequency index; RESs - Renewable energy sources; NDERs - Non-utility Distributed Energy Resource; UDERs - Utility Distributed Energy Resource; DRPs - Demand Response program Providers; ICES - Integrated community energy system; WM - Wholesale electricity market; ENS - Energy not served; DGO - Distributed generation operator; AC - Alternating Current; MCS - Monte Carlo Simulation; DTR - Dynamic Thermal Rating; V2G - Vehicle-to-grid; V2GCS - V2G charging stations; OLTC - On-load tap changer; SOPs - Soft open points; SVCs - Static var compensators

and the short-term operation conditions of ADN. The power reliability indices calculated as part of the objective function are the value of lost load and the annual outage cost at each load node determined by outage hours at each planning stage. It is noteworthy to point out that the model is time-consuming when applied to the ADN planning with hundreds of nodes.

III. ACTIVE NETWORK MANAGEMENT TECHNOLOGIES AND SERVICES IN THE ADNEP MODELS

A. DYNAMIC THERMAL RATING

Dynamic thermal rating represents a shift from static, conservative asset ratings toward real-time capacity assessment of power system components such as overhead lines, underground cables, and transformers by incorporating actual thermal and environmental conditions. Rather than relying solely on worst-case design assumptions, DTR systems monitor or forecast factors such as conductor or cable temperature, ambient temperature, wind speed, soil moisture (for underground cables), and oil or winding temperature (for transformers) to dynamically determine how much load the equipment can safely carry. Recent studies indicate that DTR can increase usable capacity by approximately 15–30% compared to static ratings and is increasingly being applied across transmission, distribution, and transformer assets [119], [120], [121]. By enabling higher utilization of existing infrastructure, DTR provides a cost-effective means to alleviate thermal bottlenecks, integrate variable renewable generation, and defer infrastructure upgrades.

In the reviewed literature that analyzed DTR as an active network management solution, three primary subsets were investigated: Dynamic line rating (DLR) for overhead conductors, DTR of underground cables, and DTR of transformers [77], [103]. For example, in publication [15] through DLR the real-time hosting capacity of Öland network, in Sweden is calculated which can improve the available hosting capacity of the network assets during favorable weather conditions. It is worth noting that in [15] two other ANM technologies and services were proposed reactive power control and power curtailment to ensure that when enhanced hosting capacity from DLR is not available, there is a solution for control of generators via curtailment.

Furthermore, the papers presented by [59] and [103] explored the application of DLR as a smart-grid-based ANM solution to enhance the efficiency and resilience of distribution networks. In [59] DLR was integrated within a strategic distribution network planning framework as an investment

alternative to conventional reinforcement, where real-time weather-dependent ampacity adjustment was used to increase feeder capacity, defer upgrades, and improve renewable generation accommodation. In contrast, [103] proposed a convex multi-objective optimization model for resilient micro-grid-oriented expansion planning, formulating a cost-effective trade-off between line hardening and DLR installation under high-impact outage scenarios. In their model, DLR dynamically updated line ratings based on wind speed, ambient temperature, and solar radiation, enhancing system resiliency above 92% while minimizing investment, operation, and emission costs.

On the other hand, publications [76], [77] extended the application of DTR from transmission lines to distribution network planning, demonstrating its effectiveness as an ANM solution. The authors of [76] proposed a stochastic ADNEP model incorporating the DTR of underground cables and transformers, where thermal models were embedded to dynamically update equipment ratings based on real-time thermal and environmental conditions. This approach captured the coupling effects of temperature and loading over time, achieving up to 15% investment cost reduction compared with static ratings. Similarly, [77] developed a bi-level distribution network planning model that jointly considered DTR of cables and transformers along with EVs active response. Using a hybrid Prim–PSO optimization framework and second-order cone relaxation, DTR was applied to adapt cable and transformer capacity under varying meteorological conditions, resulting in a 4.8% investment cost saving and a 96% increase in supply capacity.

Together, these studies demonstrate that integrating DTR into distribution network planning enables smarter asset utilization, supports renewable and EV integration, improved adaptability under varying operational and environmental conditions, and defers costly infrastructure reinforcement.

B. VOLTAGE CONTROL

The main problems relating to voltage regulation in ADNs are reverse power flow, reactive power injection that arises due to underground cabling in urban areas, shortened voltage support with ever-fewer synchronous generators due to the decommissioning of thermal diesel engines and voltage fluctuations due to intermittency of the renewable sources of the DERs. The Volt/VAr control (VVC) concept has been developed to improve the voltage quality, minimize active power losses, and maintain the voltage profile in ADNs

[122]. Voltage control devices used in ADNs to maintain voltage fluctuations include DERs, CBs, on-load tap changer, voltage regulator (VR), smart inverters (SIs), SVCs, static synchronous compensators, soft open points, and control algorithms which maintain the voltage profile, and reduce active power losses [122], [123].

To ensure voltage control in ADNs, control strategies and optimization methods have been developed and implemented, i.e. centralized control, decentralized control, distributed control and coordinated control. Detailed overviews of voltage control in active distribution networks are presented by [122], [123], and [124]. The reviews provide valuable insight into existing concepts on voltage control in ADNs, models and methods aimed at dealing with control architecture, modelling, and coordination strategies. On the other hand, in the context of ADNEP models that implemented voltage control as ANM schemes the coordinated voltage control is widely used in references, as summarized in table 3.

1) COORDINATED VOLTAGE CONTROL

For effective mitigation and suppression of voltage fluctuations in digitized ADNs, coordinated voltage control and regulation are key. To maintain the voltage profiles within the prescribed limits, discrete and continuous regulating devices need to be coordinated over a variety of timescales. Slow discrete devices such as OLTCs, CBs, and automatic voltage regulators are scheduled hourly on a day-ahead OPF-based optimization for finding the optimal setpoints of the slow time scale devices to handle the potential (predictive) voltage violations based on the prediction of renewable energy sources outputs and load demand variations [122]. On the other hand, during real-time operation, once probabilistic voltage violation is detected in the network, the reactive power of fast-responding continuous devices such as SVCs [73] and DER's smart inverter are cooperatively coordinated immediately to provide fast corrective control [122]. At the planning stage, however, it is common for studies to model the reactive capabilities of these devices as continuous variables for tractability [59], [99], [111], even though reactive power compensation equipment such as shunt capacitors, reactors, OLTC taps, and SVCs inherently operate in discrete steps representing a notable simplification of real operational behavior.

In Ref. [23] the AND planning model was created and tested for different cases, combining ANM schemes: reactive power control, active power curtailment, shunt compensation, and coordinated voltage control. For simplicity, the utility is considered the operator of the distribution network, the energy supplier to end users, and the owner of the DG. In the formulation of the model the ANM are defined as restrictions, and the cost of the voltage control devices is included in the objective function. It is worth mentioning that the authors' in [23] preferred to underestimate the capability of non-dispatchable DG units to compensate for reactive power when the reactive power control scheme is enabled.

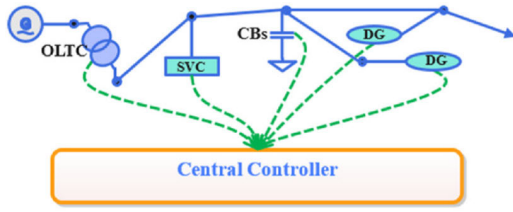
Authors in publications [56], [75] considered the OLTC as main element in the CVC scheme where the secondary voltage of the OLTC was a variable maintaining its value within a certain range. Specifically, reference [75] assumes that distribution network operators are capable to set the power factor for renewable energy resources inverters to provide reactive power support to the grid. The findings presented in [75] further highlight that the multi-DG configurations under ANM schemes could increase the potential of wind and solar power penetration at certain locations.

C. REACTIVE POWER CONTROL

Reactive power plays an important role in power grid, particularly power grid voltage management and stability [125]. Therefore, it is imperative to maintain and manage adequate reactive power reserve to ensure a stable and reliable power grid [125], [126]. In ADNs the power electronic converters interfaced renewable energy generators together with the voltage control devices can compensate for reactive power and control schemes (e.g. centralized and decentralized, see Fig 11.) are essential for reactive power management. In Sarkar et al. [125] a critical review of reactive power management in renewable rich power grids, with special emphasis on grid-codes, renewable generator capabilities, reactive power support devices, control strategies, coordination and optimization algorithms is presented. One of the key findings of their publication highlights that the implementation of centralized schemes is expensive, and hence distributed schemes should be used for better coordination of various reactive power support devices in the power network.

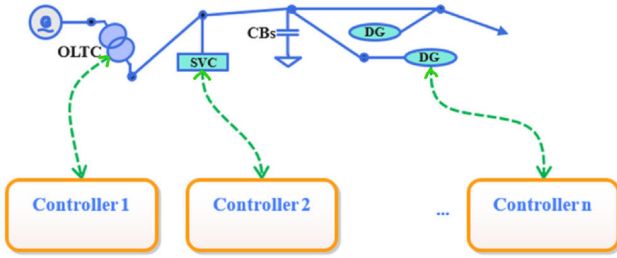
Centralized control schemes in active distribution systems are control architectures in which a central controller collects real-time measurements from across the network and computes coordinated control actions such as curtailing distributed generation, adjusting OLTCs, or managing reactive power compensators to optimize system performance. These schemes can be either rule-based, using predefined actions derived from historical data, or optimization-based, employing advanced algorithms to determine system-wide optimal settings. While offering a comprehensive view and high control accuracy, centralized approaches rely heavily on robust communication infrastructure and may face challenges related to scalability and resilience [23], [48]. By contrast, decentralized control schemes operate based on local measurements and predefined control rules without requiring communication with other network elements or a central controller. These schemes are simple, scalable, and robust to communication failures but may lead to suboptimal system-wide outcomes [23], [124].

A total of 13 research publications out of the 50 collected for this study carry out the reactive power control as an ANM in the expansion planning model. Around 70% of the articles that implemented reactive power control considered the flexibility of smart inverters to support autonomous voltage control and the capability for reactive power support [23],



a) Centralized Control architecture

Sub-station
Communications



b) Decentralized Control architecture

FIGURE 11. Examples of control architectures (modified from [122]).

[56], [58], [74], [75], [100], [109]. Usually, smart inverters have limited reactive power capacity and are therefore oversized in rating such that the active power capacity is designed to be smaller than their apparent power rating to utilize their fast and continuous reactive power support capabilities [122], as shown in Fig. 12. use to optimize the use of reactive power, the DSO tries to minimize the reactive power in the planning stage, while keeping the power factor of DGs close to one. By doing so, the DSO maintains more flexibility for the operation stage, where potential real-time control may be needed.

The papers presented by [58], [74], and [109] it is assumed that DG units can absorb or inject reactive via their inverter interface to the network, as shown in Fig. 12. Thus, DG units can operate under active power factors. Furthermore, all the variables associated with the ANM are considered as continuous variables. The authors of these papers consider the inverter apparent power to constrain the reactive power output of DG units, as illustrated in (8).

$$P_{SI}^2 + Q_{SI}^2 \leq S_{SI}^2 \quad (8)$$

In the other hand, publication [75] determined the power factor for renewable energy sources inverters to provide reactive power support to the grid. In their work the regulation used from the DSO in United Kingdom is applied and the power factor of wind turbines and PVs are kept in the range of 0.95 leading and 0.95 lagging as a constraint of the model.

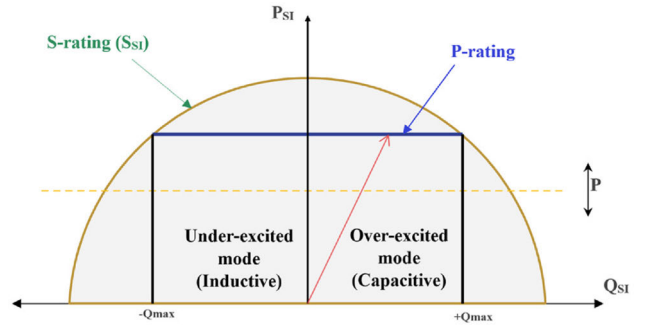


FIGURE 12. Smart inverter capacity curve [122].

D. ACTIVE POWER CURTAILMENT

The active power curtailment (APC) strategy is implemented to cut off the excessively generated active power from the supply side to tackle the overvoltage problems without influencing the power flow directions and to sustain the energy balance between the generation-side, and customers [127]. APC can be classified into two types: hard curtailment and soft curtailment [128]. In soft methods, the active power is curtailed partially according to the overvoltage deviations. In contrast, the hard methods curtail the active power completely once the voltage performance indices are violated [127]. Several studies incorporated APC into the formulation of the ADN expansion planning problem [48], [58], [59], [74], [97], [100], [109], [111]. However, curtailing the active power output of renewable DG units should be treated as a last resort and applied with caution. This is because such curtailment can have financial implications, potentially requiring partial or full compensation to DG owners depending on the prevailing regulatory framework [58], [74], [109].

Active distribution expansion planning models considering APC generally implement this ANM solution as a constraint that can be formulated as follows [100], [129].

$$(1 - c_{curt}) P_{DG}^{max} \leq P_{DG}^{curt} \leq P_{DG}^{max} \quad (9)$$

where $c_{curt} \in [0, 1]$ defines the allowed curtailment share.

It is important to note that, in most of the reviewed literature, the cost associated with APC is explicitly included in the objective function of the optimization model. For instance, in [48], the cost of curtailing active power is set significantly higher than the cost associated with reactive power control. This modeling choice reflects a deliberate prioritization of reactive power management over APC, based on the premise that reactive power control is generally a more cost-effective ANM strategy.

E. NETWORK RECONFIGURATION

Active distribution network reconfiguration (ADNR) can be defined as the dynamic process of modifying the topology of a radial distribution network through the coordinated operation of sectionalizing and tie switches to optimize system

performance [130]. This technique is modeled as a multi-objective, multi-constraint optimization problem that seeks to achieve several operational goals, including minimization of active power losses, mitigation of line and transformer overloads, enhancement of voltage profiles, improvement of power quality, and increased hosting capacity for DG. ADNRR also plays a significant role in enhancing the overall reliability and resilience of the distribution system [130], [131].

By strategically opening and closing switches to rearrange power flow paths, ADNRR enables effective load balancing across feeders, which prevents localized congestion, reduces the risk of outages, and maximizes network utilization. Additionally, it supports voltage stability, reduces the frequency and duration of switching operations, and facilitates faster system restoration following disturbances. Given its adaptability and operational benefits, ADNRR is considered a cornerstone technology in the advancement of smart grids and the integration of renewable energy sources [132], [133].

Several recent studies have emphasized the role of network reconfiguration as a crucial flexibility mechanism in the planning and operation of active distribution systems. In [18] the authors proposed a stochastic system-of-systems architecture for the adaptive expansion of smart distribution grids. Within this framework, network reconfiguration is modeled as an operational adaptability tool that allows the grid to modify its topology in response to uncertainties in demand and renewable generation. By coordinating reconfiguration with other subsystems such as DG and DR, the model enhances both the resilience and efficiency of the distribution network over time. In a related context, [51] developed a stochastic two-stage expansion planning model that jointly integrates DR and network reconfiguration. The first stage uses Monte Carlo simulations based on realistic scenarios of load, EV penetration, and renewable generation to identify potential network constraints, while the second stage applies linear programming for DR deployment and exhaustive search for optimal network reconfiguration to mitigate those constraints. This approach provides a cost-effective planning strategy that minimizes reinforcement needs by combining operational flexibility and uncertainty modeling.

Complementarily, publication [61] presented a medium-term planning framework that integrates voltage-dependent load modeling, CO₂ emission reduction, and network reconfiguration within a mixed-integer nonlinear programming (MINLP) formulation. Network reconfiguration in this model serves to minimize system losses and operational costs while maintaining radiality, with the voltage-dependent load representation enabling the optimization process to capture realistic variations in demand due to voltage changes. Additionally, the inclusion of CO₂ emission factors supports environmentally conscious planning decisions.

More recently, [107] proposed a neighborhood search-based metaheuristic for the optimal operational planning of distribution systems, where reconfiguration is realized through a branch-exchange mechanism embedded within a

MISOCP model. This algorithm iteratively explores neighboring radial configurations by closing and opening switches, using reduced sub-models to limit computational effort and a tabu list to prevent redundant searches. By integrating reconfiguration with the coordinated operation of capacitor banks and energy storage devices, the method ensures efficient power flow management and improved operational performance.

Overall, these studies highlight that incorporating network reconfiguration through stochastic, mathematical, or heuristic optimization significantly enhances system adaptability, cost efficiency, and environmental sustainability in modern distribution network planning.

F. ANALYTICAL REVIEW AND KEY FINDINGS ON ANM INTEGRATION IN EXPANSION PLANNING MODELS

In this survey, a total of 50 research works published over the past decade are analyzed. Among these, in 27 studies ANM is implemented either explicitly under the label active network management or active distribution or implicitly via detailed models of flexible devices and services. Specifically, 5 of these works employed distribution network reconfiguration, 12 implemented APC, and another 13 utilized reactive power control. Additionally, 4 studies incorporated DTR, while 8 publications applied coordinated voltage control, as summarized in Fig. 13. ANM technologies are generally integrated into the ADNEP formulation via operational constraints and cost terms, allowing planners to capture both short-term flexibility and long-term system benefits. Their mathematical modeling frequently employs MILP, MINLP, or MISOCP, depending on the linearization of network equations.

Within the objective function, ANM is predominantly captured through cost components, such as APC costs [100], [111]. Additionally, several studies account for ANM-related expenses as part of the investment costs, particularly for devices involved in Volt/Var control, including CBs, OLTCs, and SVCs [23], [61], [73]. This dual representation of costs underscores the multifaceted role of ANM in balancing the trade-offs between operational efficiency and strategic capital investments in distribution networks.

Table 3 provides a comprehensive summary of the studies that integrated ANM solutions within the scope of active distribution network expansion planning. The table presents key attributes of each study, including the adopted planning horizon, the types of Volt/Var control devices, and the specific ANM schemes considered.

Across the reviewed ADNEP and ANM studies, the mathematical treatment of network management schemes varies significantly, balancing realism and computational tractability. ANM schemes are implemented using a combination of continuous control variables, discrete installation or switching decisions, and linear or convexified power-flow constraints embedded directly in the planning formulation.

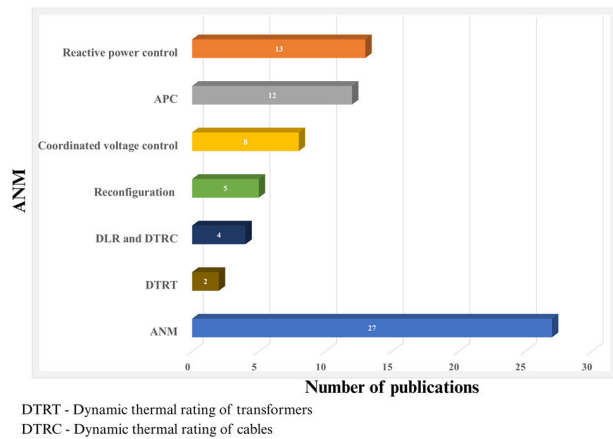


FIGURE 13. Types of ANM technologies or services implemented in the ADNEP models.

Active distribution network reconfiguration is almost universally represented through binary variables that indicate the open or closed status of sectionalizing and tie switches. This modeling makes reconfiguration a discrete, combinatorial problem embedded in either MILP or MINLP formulations. To maintain tractability, several authors [58], [74], [75], [107] adopt linearized power-flow models, which enable the use of integer-linear optimization techniques. In these cases, reconfiguration directly affects both voltage and loss constraints, and indirectly the objective function by minimizing operational costs or energy losses. Conversely, a few studies [59], [63] maintain nonlinear AC formulations to capture detailed voltage behavior, at the expense of scalability.

Voltage regulation devices such as OLTCs and capacitor banks are typically modeled through discrete control positions in operational studies but approximate as continuous variables in planning. The continuous modeling approach, adopted in works such as [59], [98], [99], and [114], simplifies computation by replacing stepwise tap ratios or switched reactive steps with continuous control intervals. This converts otherwise mixed-integer formulations into convex or linear subproblems. However, such simplifications overestimate controllability in operation, as practical devices have discrete and time-delayed switching. When explicitly included, the cost of voltage regulation appears in the objective function either as switching penalties or through loss reduction benefits, depending on the modeling perspective.

In more recent ANM formulations [74], [94], [113], smart inverters reactive power control is modeled as a continuous, convex, and fast-response ANM variable that operates within the inverter's apparent power capability. Unlike discrete voltage devices, inverter-based VAR support enables finer and near-real-time voltage regulation, making it one of the most analytically tractable and operationally realistic ANM options. However, most studies still assume central control and neglect communication delays, inverter drop nonlinearity, and coordination with other voltage devices,

simplifications that can overestimate practical flexibility potential.

On the other hand, DG curtailment, ESSs, and DR are generally represented by continuous power or energy variables constrained by physical or contractual limits. DG curtailment enters as a continuous active power reduction variable, usually penalized in the objective function to reflect lost generation or compensation payments. ESS is modeled as a continuous intertemporal variable pair (charge/discharge power and energy balance) with linear constraints capturing efficiency and state-of-charge.

More recent works such as [76], [94], and [103] model DTR and SOPs as continuous network flexibility variables. DTR modifies current or power-flow constraints dynamically based on ambient conditions (nonlinear but convexified in many cases), while SOPs introduce linear power-coupling constraints between feeders. Both contribute directly to loss minimization and capacity deferral objectives.

An examination of Table 3 indicates that many studies employed standardized medium-voltage distribution networks (typically operating between 12.66 kV and 33 kV and comprising 16 to 119 buses) to assess and benchmark the proposed models. This practice enhances comparability across methodologies. In addition, eleven studies applied their models to real-world distribution systems, thereby improving practical relevance. Notable examples include the medium-voltage distribution network of Grenoble, France [51], consisting of 136 substations at 20 kV, and the Colombian distribution system analyzed in [61], which comprises 2 307 load nodes, 2 335 branches, and six substations, with a nominal voltage of 14.4 kV. Finally a distribution network located near the Cambridge area in the United Kingdom was shown in [59] where the methodology proposed has been tested on a real 33 kV distribution network with 29 buses and 36 branches.

In terms of planning horizons, long-term planning horizons dominate literature, with thirteen publications adopting planning periods beyond ten years, eight addressed mid-term planning, while six papers do not explicitly state any planning horizon. Notably, twelve articles [18], [48], [51], [61], [63], [76], [77], [94], [95], [103], [107], [113] do not clearly state whether ANM technologies or services were incorporated into the ADNEP models. Although these studies implicitly integrate ANM schemes, as reflected in Table 3, the absence of explicit identification hinders consistent classification and limits the ability to conduct a systematic comparative analysis across literature.

Among the reviewed studies that incorporated ANM technologies and services into expansion planning models, more than 55% employed a combination of two or more ANM strategies. This trend underscores the growing recognition of the value of integrated approaches in effectively addressing the escalating complexity of modern power systems. Despite this progress, the prevailing modeling paradigm in literature continues to rely predominantly on single-objective optimization techniques.

TABLE 3. Summary of reviewed studies implementing ANM solutions in ADNEP models.

Ref.	Test Network		Planning horizons (Years)	Volt/VAr control devices	ANM technologies or services					
	Standard	Real			DTR	DLR	CVC	APC	ADNR	RPC
[106]	Y		5	SI & OLTC			✓		✓	
[18]	Y	Y	4						✓	
[105]	Y		5			✓				
[52]		Y	15						✓	
[98]	Y			OLTC			✓			
[100]	Y		15							
[101]	Y		15	ESS						
[49]	Y		10	SI				✓		✓
[111]	Y		10	SI				✓		✓
[75]	Y	Y	20	SI				✓		✓
[59]	Y	Y	20	SI				✓		✓
[23]	Y		20	SI, OLTC, CBs & SVC			✓	✓		✓
[116]		Y		SVC, OLTC & SOP			✓	✓		✓
[78]	Y		5		**					
[62]	Y	Y	5	CBs & VR					✓	
[60]		Y	16	SVC		✓		✓		
[57]	Y		3	SI & OLTC			✓			✓
[77]	Y			SI & OLTC			✓			✓
[96]	Y		5	SI, OLTC & SOP			✓		✓	✓
[79]	Y		5		**					
[115]	Y	Y		SI, OLTC, CBs & SVC			✓			✓
[102]	Y	Y	10	SI, OLTC, CBs & SVC				✓		✓
[113]	Y		10	OLTC & SVC				✓		
[97]	Y		20					✓		
[109]	Y			CB					✓	
[99]	Y		10					✓		✓
[64]	Y			CBs & SVC				✓		

** DTR - Dynamic thermal rating of cables and transformers; Y – Yes

Notably, a growing number of studies, [18], [56], [58], [77], [94], [95], [97], [103], [111], [113], and [114] have investigated more advanced modeling paradigms, including multi-objective, bilevel and tri-level optimization. These formulations provide a more holistic framework for capturing the trade-offs among conflicting operational objectives and the hierarchical nature of decision-making processes inherent in real-world power systems.

Moreover, around 70% of ANM-based expansion planning models directly addressed uncertainties parameters. Of these, approximately 53% employed stochastic optimization methods to account for uncertain parameters, such as variability in renewable energy generation and fluctuations in electricity demand. This observation reveals that ADNEP models are moving from passive reinforcement planning toward ANM-enabled and uncertainty-aware methodologies.

Nevertheless, across the 27 ANM-enabled ADNEP studies the implementation, communication, and control-system

costs are consistently omitted in optimization models, despite their central role in enabling ANM deployment. In almost all cases, the planning frameworks account only for device-level capital costs such as SVCs, ESS, DLR equipment, SOPs, OLTC upgrades, or reactive power resources, while assuming that the underlying information and communication technology (ICT), monitoring, automation, and decision-support infrastructure is available at no additional cost. For instance, [59], [100] model the investment costs of smart grid technologies and their operational benefits, yet they omit expenditures related to sensors, communication links, or advanced control platforms required for real-time ANM coordination. Similar patterns are observed in [94], where SOP, DER, and operational costs are captured in detail but the optimization implicitly presumes an existing ANM-capable ICT infrastructure. Even studies that acknowledge the need for ANM support systems, such as [109], represent this requirement only through abstract cost terms rather than explicit

modelling of real implementation expenditures. In [23], the authors explicitly note this gap, stating that further work is required to assess the cost of implementing ANM strategies and deploying the dedicated infrastructure they rely on. Taken together, these findings validate that implementation costs remain largely absent from existing ADNEP–ANM models, underscoring a critical research gap with direct implications for techno-economic realism and practical DSO decision-making.

IV. DISCUSSIONS AND FUTURE RESEARCH DIRECTIONS

Although this manuscript compiles and synthesizes findings from fifty peer-reviewed studies on active distribution network planning (Tables 1 and 2), the analysis presented in this discussion section focuses specifically on the 27 articles that incorporate ANM functionalities within their formulations. Concentrating on this subset allows a more coherent assessment of the modelling approaches, assumptions, and solution techniques used for ANM-enabled planning. It also provides a clearer basis for evaluating the practical applicability of these methods in the European context, where DSOs operate under unbundling rules and procure flexibility through market-based mechanisms rather than through direct ownership of DER assets.

The reviewed ADNEP–ANM literature shows remarkable methodological progress over the last decade, yet several structural misalignments remain between academic models and real DSO planning practice. Several papers make strong contributions, particularly in demonstrating the technical and economic benefits of ANM schemes and in recognizing the importance of uncertainty modelling. However, the review also reveals important gaps that must be addressed to make these methods suitable for European distribution system planning.

A. STRENGTHS AND POSITIVE DEVELOPMENTS IN THE LITERATURE

Many studies provide compelling evidence that ANM can significantly enhance hosting capacity, reduce curtailment, and defer traditional reinforcement [63], [74], [75], [100], [109]. A wide set of ANM options (e.g. DG reactive power control, OLTC coordination, DTR, and EV-based flexibility) has been shown to reduce network stress and improve operational security when appropriately coordinated [23]. Literature also increasingly acknowledges that deterministic peak-case planning is insufficient; stochastic models, scenario-based planning, and probabilistic forecasts capture future uncertainties far more accurately [61], [104], [107]. Recent multistage and bilevel formulations demonstrate growing awareness of the temporal separation between long-term infrastructure decisions and short-term operational feasibility [77], [95], [114].

Furthermore, a few papers take meaningful steps toward modeling the distribution system as a system of systems. Arasteh et al. [18] conceptualize distribution network operation as an adaptive multi-agent environment operating

under uncertainty, and Zeng et al. [97] explicitly model the economic interaction between DSOs, DG owners, and the operator in a tri-level framework. Even if computationally demanding, these contributions highlight the structural importance of stakeholder interactions.

B. LIMITATIONS AND CONCEPTUAL MISALIGNMENTS

Despite these advances, several limitations are repeatedly observed across the literature.

First, many papers continue to optimize the siting and sizing of DERs as if these were DSO-controlled assets. This contradicts European regulatory unbundling, where DSOs cannot own generation or storage assets except in exceptional circumstances. These models remain technically insightful, but their results should be interpreted as hosting-capacity explorations or locational signal analyses, not as literal DSO planning recommendations.

Second, most models treat flexibility as physical DER capacity, rather than as market-based services such as congestion management, voltage support, peak shaving, or restoration services. In practice, DERs may simultaneously provide multiple services to multiple markets, and these services are often competing. The absence of service-based modelling and resource stacking means most studies implicitly overestimate the amount of flexibility available to the DSO.

Third, although several papers reference prosumers, DR, aggregators, or EV users, almost none model these stakeholders as independent decision-makers. Instead, a single central planner optimizes everything through a composite objective function. While this is mathematically convenient, it diverges sharply from European electricity market design, where aggregators optimize their own portfolios and DSOs must procure flexibility via regulated processes. Only a small number of studies capture strategic interactions between actors, but these are generally too computationally demanding to scale.

Finally, engineering realism is often sacrificed for optimization elegance. Many models treat inherently discrete devices (e.g. OLTC taps, capacitor banks, reactors) as continuous variables, omit ITC, simplified AC power-flow models or protection constraints and rely on optimistic assumptions regarding forecast quality. While these assumptions improve tractability, they must be carefully acknowledged to avoid misleading conclusions.

C. IMPLICATIONS FOR DSO

The literature makes it clear that there is no global optimum that simultaneously maximizes welfare for DSOs, aggregators, and prosumers. Distribution grid planning is fundamentally an engineering activity constrained by regulation, operational security, cost recovery rules, and practical implementation considerations. Optimization models are best viewed as decision-support tools rather than decision-makers.

Several methods identified in the literature are sufficiently mature for immediate or near-term use by DSOs as decision-support tools, including stochastic expansion planning,

ANM-enhanced planning with convex relaxations, and multistage reinforcement analysis. More advanced bilevel or game-theoretic models, present high potential but currently serve best as research explorations rather than operational planning instruments.

A key message is therefore that **future ADNEP research must move toward models that:**

1. respect the DSO's actual role under European regulatory principles,
2. treat DER-driven flexibility as a service, not an asset,
3. acknowledge multi-stakeholder decision processes, and
4. retain engineering realism while embracing uncertainty.

This shift will enable planning models that are more technically consistent, economically grounded, and implementable.

D. FUTURE RESEARCH DIRECTIONS

Building on the insights obtained from the fifty studies reviewed in this manuscript, as well as the deeper analysis of the 27 articles that incorporate ANM within their planning formulations, several structural and recurring research gaps emerge. Although some individual studies may have begun addressing parts of these challenges, the evidence across the full body of literature shows that these issues persist at the field level and therefore require coordinated attention. The research questions below highlight high-impact research directions that follow directly from the limitations identified:

Research question 1: How can flexibility be represented as multi-service, competing products within ADNEP?

- Formulate flexibility procurement as multi-service resource allocation.
- Model service stacking (congestion management, voltage support, arbitrage).
- Incorporate aggregator strategies and multi-market participation into planning models.

Research question 2: How should ADNEP reflect European DSO constraints and market rules?

- Incorporate cost recovery, tariff limitations, and neutrality constraints consistently with regulatory frameworks.
- Model DERs decisions as market-driven choices by independent actors rather than as DSO-controlled assets.

Research question 3: How can multi-actor behavior be represented realistically but tractably?

- Develop reduced-form models of aggregator/prosumer response.
- Explore approximate game-theoretic or learning-based stakeholder models.

Research question 4: How to integrate engineering realism (discrete devices, ICT limits, protection, time granularity) into computationally tractable models?

- Develop hybrid AC–SOCP or related formulations that capture discrete device behavior where necessary.
- Create lightweight but realistic ANM technologies and services models.

- Represent observability, communication requirements, and other ICT constraints that influence the feasibility of ANM deployment.

Research question 5: What is the minimal modelling complexity required for reliable planning?

- Identify which uncertainties matter most.
- Determine the value of additional ANM levers under different network types.

Research question 6: How should DSOs evaluate ANM-enabled investments under long-term uncertainty?

- Combine stochastic, adaptive, and robust elements into hybrid planning frameworks.
- Explicitly consider regulatory, societal, and implementation uncertainty.
- Include the full cost of implementation, communication, control platforms, and monitoring infrastructure that enable ANM functionality.

Research question 7: How can ADNEP adopt modular and composable modelling structures to enable scalable integration of ANM?

- Develop interoperable, modular model components for representing ANM technologies and services.
- Establish composable modelling frameworks that allow ANM functionalities to be integrated consistently across different network sizes, planning horizons, and technological contexts.
- Identify abstraction and model-reduction techniques that preserve essential ANM behavior while maintaining tractability in large-scale ADNEP formulations.

Research question 8: How do ANM strategies influence equipment life-cycles and asset-management decisions?

- Assess how ANM-driven operational patterns affect asset ageing, reliability, and maintenance requirements.
- Evaluate the implications for asset-replacement timing and long-term renewal strategies.
- Compare life-cycle economic outcomes of conventional reinforcement and ANM-based solutions.

V. CONCLUSION

This review synthesizes recent developments in ADNEP–ANM modelling and evaluates their suitability for the evolving European distribution system environment. The state of the art shows a clear trajectory: from deterministic, static optimization toward multistage, uncertainty-aware, and hybrid modelling frameworks. These models increasingly incorporate ANM schemes, flexibility resources, and operational feasibility considerations. Yet, despite impressive technical depth, many academic formulations remain misaligned with regulatory constraints and real DSO responsibilities.

Three overarching conclusions emerge.

First, ANM and flexibility are essential components of modern ADNEP. Their coordinated use demonstrably reduces reinforcement needs and improves resilience. However, the physical representation of DERs in most

models must be replaced by service-based flexibility products that align with European market design.

Second, multi-stage optimization is not optional, future planning tools must reflect the temporal separation between investment decisions, flexibility procurement, and real-time operation. While multistage and bilevel formulations already exist, they must be redesigned to reflect actual market roles, actor autonomy, and procurement processes.

Third, no single “optimal” plan exists when multiple actors with conflicting incentives are considered. Engineering judgement must remain central: optimization tools should support, not replace, DSO’s technical expertise. Transparent, explainable, and regulation-compliant models are far more valuable than theoretically perfect ones.

In summary, the next generation of ADNEP models must be multi-stage, multi-actor, service-based, uncertainty-aware, and engineering-grounded. They should balance mathematical rigor with institutional realism and pave the way for planning frameworks that DSOs can implement within European regulatory constraints.

REFERENCES

- [1] *Grids, the Missing Link. An EU Action Plan for Grids*, European Commission, Brussels, Belgium, 2023.
- [2] EU DSO Entity. (2025). *DSOs in the Energy Transition*. Accessed: Nov. 1, 2024. [Online]. Available: <https://eudsoentity.eu/dsos-in-the-energy-transition>
- [3] T. D. de Lima, F. Lezama, J. Soares, J. F. Franco, and Z. Vale, “Modern distribution system expansion planning considering new market designs: Review and future directions,” *Renew. Sustain. Energy Rev.*, vol. 202, Sep. 2024, Art. no. 114709, doi: [10.1016/j.rser.2024.114709](https://doi.org/10.1016/j.rser.2024.114709).
- [4] K. H. M. Azmi, N. A. M. Radzi, N. A. Azhar, F. S. Samidi, I. T. Zulkifli, and A. M. Zainal, “Active electric distribution network: Applications, challenges, and opportunities,” *IEEE Access*, vol. 10, pp. 134655–134689, 2022, doi: [10.1109/ACCESS.2022.3229328](https://doi.org/10.1109/ACCESS.2022.3229328).
- [5] G. Varathan and J. B. Edward, “A review of uncertainty management approaches for active distribution system planning,” *Renew. Sustain. Energy Rev.*, vol. 205, Nov. 2024, Art. no. 114808, doi: [10.1016/j.rser.2024.114808](https://doi.org/10.1016/j.rser.2024.114808).
- [6] Y. Xiang, J. Liu, F. Li, Y. Liu, Y. Liu, R. Xu, Y. Su, and L. Ding, “Optimal active distribution network planning: A review,” *Electric Power Compon. Syst.*, vol. 44, no. 10, pp. 1075–1094, Jun. 2016, doi: [10.1080/15325008.2016.1156194](https://doi.org/10.1080/15325008.2016.1156194).
- [7] IRENA. (Feb. 2019). *Innovation Landscape Brief: Future Role of Distribution System Operators*. [Online]. Available: https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2019/Feb/IRENA_Landscape_Future_DSOs_2019.pdf
- [8] G. Celli, “A comparison of distribution network planning solutions: Traditional reinforcement versus integration of distributed energy storage,” in *Proc. IEEE Grenoble Conf.*, Jun. 2013, pp. 1–6, doi: [10.1109/PTC.2013.6652338](https://doi.org/10.1109/PTC.2013.6652338).
- [9] O. Samuelsson, S. Repo, R. Jessler, J. Aho, M. Karenlampi, and A. Malmquist, “Active distribution network—Demonstration project ADINE,” in *Proc. IEEE PES Innov. Smart Grid Technol. Conf. Eur. (ISGT Eur.)*, Oct. 2010, pp. 1–8, doi: [10.1109/ISGTEUROPE.2010.5638988](https://doi.org/10.1109/ISGTEUROPE.2010.5638988).
- [10] P. Wiest, D. Gross, K. Rudion, and A. Probst, “Rapid identification of worst-case conditions: Improved planning of active distribution grids,” *IET Gener., Transmiss. Distrib.*, vol. 11, no. 9, pp. 2412–2417, Jun. 2017, doi: [10.1049/iet-gtd.2017.0148](https://doi.org/10.1049/iet-gtd.2017.0148).
- [11] D. Kuiken and H. F. Más, “Integrating demand side management into EU electricity distribution system operation: A Dutch example,” *Energy Policy*, vol. 129, pp. 153–160, Jun. 2019, doi: [10.1016/j.enpol.2019.01.066](https://doi.org/10.1016/j.enpol.2019.01.066).
- [12] W. Medjroubi, U. P. Müller, M. Scharf, C. Matke, and D. Kleinhans, “Open data in power grid modelling: New approaches towards transparent grid models,” *Energy Rep.*, vol. 3, pp. 14–21, Nov. 2017, doi: [10.1016/j.egyr.2016.12.001](https://doi.org/10.1016/j.egyr.2016.12.001).
- [13] R. Li, W. Wang, Z. Chen, J. Jiang, and W. Zhang, “A review of optimal planning active distribution system: Models, methods, and future researches,” *Energies*, vol. 10, no. 11, p. 1715, Oct. 2017, doi: [10.3390/en10111715](https://doi.org/10.3390/en10111715).
- [14] CIRED. *Network Planning & System Design With Flexibility*. Accessed: Apr. 2, 2025. [Online]. Available: <https://www.cired.net/working-group/network-planning-system-design-with-flexibility/>
- [15] A. Dahlgren, D. Panic, T. Svensson, R. MacDonald, and F. Silver, “Increasing DER hosting capacity in Oland, Sweden, via active network management (ANM),” in *Proc. CIRED*, Jun. 2024, pp. 809–813, doi: [10.1049/icp.2024.2020](https://doi.org/10.1049/icp.2024.2020).
- [16] E. Hillberg, H. Pihl, M. Edvall, S. Hallhagen, O. Samuelsson, and M. Lundberg. (2020). *Active Network Management for All: ANM4LA Collaborative Research Project*. RISE Research Institutes of Sweden. [Online]. Available: <https://ri.diva-portal.org/smash/get/diva2:1684787/FULLTEXT01.pdf>
- [17] M. J. Ghadi, A. Rajabi, S. Ghavidel, A. Azizivahed, L. Li, and J. Zhang, “From active distribution systems to decentralized microgrids: A review on regulations and planning approaches based on operational factors,” *Appl. Energy*, vol. 253, Nov. 2019, Art. no. 113543, doi: [10.1016/j.apenergy.2019.113543](https://doi.org/10.1016/j.apenergy.2019.113543).
- [18] H. Arasteh, V. Vahidinasab, M. S. Sepasian, and J. Aghaei, “Stochastic system of systems architecture for adaptive expansion of smart distribution grids,” *IEEE Trans. Ind. Inform.*, vol. 15, no. 1, pp. 377–389, Jan. 2019, doi: [10.1109/TII.2018.2808268](https://doi.org/10.1109/TII.2018.2808268).
- [19] H. Gao, R. Wang, Y. Liu, L. Wang, Y. Xiang, and J. Liu, “Data-driven distributionally robust joint planning of distributed energy resources in active distribution network,” *IET Gener., Transmiss. Distrib.*, vol. 14, no. 9, pp. 1653–1662, May 2020, doi: [10.1049/iet-gtd.2019.1565](https://doi.org/10.1049/iet-gtd.2019.1565).
- [20] R. S. Pinto, C. Unsuhay-Vila, and F. H. Tabarro, “Coordinated operation and expansion planning for multiple microgrids and active distribution networks under uncertainties,” *Appl. Energy*, vol. 297, Sep. 2021, Art. no. 117108, doi: [10.1016/j.apenergy.2021.117108](https://doi.org/10.1016/j.apenergy.2021.117108).
- [21] X. Shen, M. Shahidehpour, Y. Han, S. Zhu, and J. Zheng, “Expansion planning of active distribution networks with centralized and distributed energy storage systems,” *IEEE Trans. Sustain. Energy*, vol. 8, no. 1, pp. 126–134, Jan. 2017, doi: [10.1109/TSTE.2016.2586027](https://doi.org/10.1109/TSTE.2016.2586027).
- [22] A. Zakariazadeh, S. Jadid, and P. Siano, “Stochastic multi-objective operational planning of smart distribution systems considering demand response programs,” *Electr. Power Syst. Res.*, vol. 111, pp. 156–168, Jun. 2014, doi: [10.1016/j.epsr.2014.02.021](https://doi.org/10.1016/j.epsr.2014.02.021).
- [23] J. Lazo and D. Watts, “Stochastic model for active distribution networks planning: An analysis of the combination of active network management schemes,” *Renew. Sustain. Energy Rev.*, vol. 191, Mar. 2024, Art. no. 114156, doi: [10.1016/j.rser.2023.114156](https://doi.org/10.1016/j.rser.2023.114156).
- [24] A. S. Al-Bukhaytan, A. T. Al-Awami, and A. M. Muqbel, “An active distribution network planning model for distributed energy resources and distribution network,” in *Proc. IEEE PES Innov. Smart Grid Technol.-Asia*, Dec. 2021, pp. 1–5, doi: [10.1109/ISGTASIA49270.2021.9715581](https://doi.org/10.1109/ISGTASIA49270.2021.9715581).
- [25] A. S. Bin Humayd and K. Bhattacharya, “Distribution system planning to accommodate distributed energy resources and PEVs,” *Electr. Power Syst. Res.*, vol. 145, pp. 1–11, Apr. 2017, doi: [10.1016/j.epsr.2016.12.016](https://doi.org/10.1016/j.epsr.2016.12.016).
- [26] B. Canizes, J. Soares, F. Lezama, C. Silva, Z. Vale, and J. M. Corchado, “Optimal expansion planning considering storage investment and seasonal effect of demand and renewable generation,” *Renew. Energy*, vol. 138, pp. 937–954, Aug. 2019, doi: [10.1016/j.renene.2019.02.006](https://doi.org/10.1016/j.renene.2019.02.006).
- [27] A. Nasri, A. Abdollahi, and M. Rashidinejad, “Multi-stage and resilience-based distribution network expansion planning against hurricanes based on vulnerability and resiliency metrics,” *Int. J. Electr. Power Energy Syst.*, vol. 136, Mar. 2022, Art. no. 107640, doi: [10.1016/j.ijepes.2021.107640](https://doi.org/10.1016/j.ijepes.2021.107640).
- [28] R. S. Pinto, C. Unsuhay-Vila, and T. S. P. Fernandes, “Multi-objective and multi-period distribution expansion planning considering reliability, distributed generation and self-healing,” *IET Gener., Transmiss. Distrib.*, vol. 13, no. 2, pp. 219–228, Jan. 2019, doi: [10.1049/iet-gtd.2018.5037](https://doi.org/10.1049/iet-gtd.2018.5037).
- [29] X. Wang, F. Li, L. Bai, and X. Fang, “DLMP of competitive markets in active distribution networks: Models, solutions, applications, and visions,” *Proc. IEEE*, vol. 111, no. 7, pp. 725–743, Jul. 2023, doi: [10.1109/JPROC.2022.3177230](https://doi.org/10.1109/JPROC.2022.3177230).

- [30] A. Ehsan and Q. Yang, "State-of-the-art techniques for modelling of uncertainties in active distribution network planning: A review," *Appl. Energy*, vol. 239, pp. 1509–1523, Apr. 2019, doi: [10.1016/j.apenergy.2019.01.211](https://doi.org/10.1016/j.apenergy.2019.01.211).
- [31] M. J. Ghadi, S. Ghavidel, A. Rajabi, A. Azizivahed, L. Li, and J. Zhang, "A review on economic and technical operation of active distribution systems," *Renew. Sustain. Energy Rev.*, vol. 104, pp. 38–53, Apr. 2019, doi: [10.1016/j.rser.2019.01.010](https://doi.org/10.1016/j.rser.2019.01.010).
- [32] G. L. Aschidamini, G. A. da Cruz, M. Resener, M. J. S. Ramos, L. A. Pereira, B. P. Ferraz, S. Haffner, and P. M. Pardalos, "Expansion planning of power distribution systems considering reliability: A comprehensive review," *Energies*, vol. 15, no. 6, p. 2275, Mar. 2022, doi: [10.3390/en15062275](https://doi.org/10.3390/en15062275).
- [33] A. Rastgou, "Distribution network expansion planning: An updated review of current methods and new challenges," *Renew. Sustain. Energy Rev.*, vol. 189, Jan. 2024, Art. no. 114062, doi: [10.1016/j.rser.2023.114062](https://doi.org/10.1016/j.rser.2023.114062).
- [34] P. S. Georgilakis and N. D. Hatzigargyriou, "A review of power distribution planning in the modern power systems era: Models, methods and future research," *Electr. Power Syst. Res.*, vol. 121, pp. 89–100, Apr. 2015, doi: [10.1016/j.epsr.2014.12.010](https://doi.org/10.1016/j.epsr.2014.12.010).
- [35] J. Liu, H. Gao, Z. Ma, and Y. Li, "Review and prospect of active distribution system planning," *J. Modern Power Syst. Clean Energy*, vol. 3, no. 4, pp. 457–467, Dec. 2015, doi: [10.1007/s40565-015-0170-7](https://doi.org/10.1007/s40565-015-0170-7).
- [36] Z. Luo, Y. Liu, and C. Wang, "Review on coordination and planning of active distribution network," in *Proc. 5th Int. Conf. Power Renew. Energy (ICPRE)*, Sep. 2020, pp. 517–522, doi: [10.1109/ICPRE51194.2020.9233134](https://doi.org/10.1109/ICPRE51194.2020.9233134).
- [37] Y. Xiang, H. Yao, J. Yang, J. Dai, and J. Liu, "Optimal planning of distribution network with transactive energy: Review and framework," *IET Smart Grid*, vol. 7, no. 2, pp. 120–129, Apr. 2024, doi: [10.1049/stg2.12114](https://doi.org/10.1049/stg2.12114).
- [38] V. Vahidinasab, M. Tabarzadi, H. Arasteh, M. I. Alizadeh, M. M. Beigi, H. R. Sheikhzadeh, K. Mehran, and M. S. Sepasian, "Overview of electric energy distribution networks expansion planning," *IEEE Access*, vol. 8, pp. 34750–34769, 2020, doi: [10.1109/ACCESS.2020.2973455](https://doi.org/10.1109/ACCESS.2020.2973455).
- [39] L. Gacitua, P. Gallegos, R. Henriquez-Auba, Á. Lorca, M. Negrete-Pincetic, D. Olivares, A. Valenzuela, and G. Wenzel, "A comprehensive review on expansion planning: Models and tools for energy policy analysis," *Renew. Sustain. Energy Rev.*, vol. 98, pp. 346–360, Dec. 2018, doi: [10.1016/j.rser.2018.08.043](https://doi.org/10.1016/j.rser.2018.08.043).
- [40] V. A. Evangelopoulos, P. S. Georgilakis, and N. D. Hatzigargyriou, "Optimal operation of smart distribution networks: A review of models, methods and future research," *Electr. Power Syst. Res.*, vol. 140, pp. 95–106, Nov. 2016, doi: [10.1016/j.epsr.2016.06.035](https://doi.org/10.1016/j.epsr.2016.06.035).
- [41] C. D'Adamo, S. Jupe, and C. Abbey, "Global survey on planning and operation of active distribution networks—Update of CIGRE C6.11 working group activities," in *Proc. 20th Int. Conf. Exhib. Electr. Distrib.*, Jun. 2009, pp. 1–4, doi: [10.1049/cp.2009.0836](https://doi.org/10.1049/cp.2009.0836).
- [42] M. B. Hadi, M. Moeini-Aghtaie, M. Khoshjahan, and P. Dehghanian, "A comprehensive review on power system flexibility: Concept, services, and products," *IEEE Access*, vol. 10, pp. 99257–99267, 2022, doi: [10.1109/ACCESS.2022.3206428](https://doi.org/10.1109/ACCESS.2022.3206428).
- [43] J. Wang, Y. Zhang, Q. Chen, Y. Cui, X. Chen, Y. Zhao, J. Ping, and Y. Zhao, "A comprehensive review of active distribution grid research: Coordinated dispatch of generation, grid, load and storage," in *Proc. 7th Int. Conf. Energy, Electr. Power Eng. (CEEPE)*, Apr. 2024, pp. 716–721, doi: [10.1109/ceepe62022.2024.10586392](https://doi.org/10.1109/ceepe62022.2024.10586392).
- [44] J. Zhao, Z. Wu, H. Long, H. Sun, X. Wu, C. Chan, and M. Shahidehpour, "Optimal operation control strategies for active distribution networks under multiple states: A systematic review," *J. Modern Power Syst. Clean Energy*, vol. 12, no. 5, pp. 1333–1344, Sep. 2024, doi: [10.35833/mpce.2023.000372](https://doi.org/10.35833/mpce.2023.000372).
- [45] N. Shaukat, M. R. Islam, Md. M. Rahman, B. Khan, B. Ullah, S. M. Ali, and A. Fekih, "Decentralized, democratized, and decarbonized future electric power distribution grids: A survey on the paradigm shift from the conventional power system to micro grid structures," *IEEE Access*, vol. 11, pp. 60957–60987, 2023, doi: [10.1109/ACCESS.2023.3284031](https://doi.org/10.1109/ACCESS.2023.3284031).
- [46] Y. Zou, Y. Xu, X. Feng, R. T. Naayagi, and B. Soong, "Transactive energy systems in active distribution networks: A comprehensive review," *CSEE J. Power Energy Syst.*, vol. 8, no. 5, pp. 1302–1317, Sep. 2022, doi: [10.17775/CSEEJPES.2021.03290](https://doi.org/10.17775/CSEEJPES.2021.03290).
- [47] S. A. A. Kazmi, M. K. Shahzad, A. Z. Khan, and D. R. Shin, "Smart distribution networks: A review of modern distribution concepts from a planning perspective," *Energies*, vol. 10, no. 4, p. 501, Apr. 2017, doi: [10.3390/en10040501](https://doi.org/10.3390/en10040501).
- [48] S. Karagiannopoulos, P. Aristidou, and G. Hug, "Hybrid approach for planning and operating active distribution grids," *IET Gener., Transmiss. Distrib.*, vol. 11, no. 3, pp. 685–695, Feb. 2017, doi: [10.1049/iet-gtd.2016.0642](https://doi.org/10.1049/iet-gtd.2016.0642).
- [49] S. Kazmi, M. Shahzad, and D. Shin, "Multi-objective planning techniques in distribution networks: A composite review," *Energies*, vol. 10, no. 2, p. 208, Feb. 2017, doi: [10.3390/en10020208](https://doi.org/10.3390/en10020208).
- [50] (Jul. 2025). *ACER and CEER, ACER-CEER Guidance on Electricity Distribution Network Planning*. [Online]. Available: <https://www.acer.europa.eu/sites/default/files/documents/Publications/ACER-CEER-electricity-distribution-network-planning-guidance-2025.pdf>
- [51] V. Gouin, M.-C. A. Herault, and B. Raison, "Stochastic integration of demand response and reconfiguration in distribution network expansion planning," *IET Gener., Transmiss. Distrib.*, vol. 12, no. 20, pp. 4536–4545, Nov. 2018, doi: [10.1049/iet-gtd.2018.5833](https://doi.org/10.1049/iet-gtd.2018.5833).
- [52] European Commission, Brussels, Belgium. (Dec. 10, 2025). *Guidance on Efficient and Timely Grid Connections, C(2025) 8473 Final*. [Online]. Available: https://energy.ec.europa.eu/document/download/62c46b3d-0df9-42a1-a5fe-c3c71ed5f18c_en?filename=C_2025_8473_1_EN_ACT_part1_v4.pdf
- [53] Q. Gemine, D. Ernst, and B. Cornélusse, "Active network management for electrical distribution systems: Problem formulation, benchmark, and approximate solution," *Optim. Eng.*, vol. 18, no. 3, pp. 587–629, Sep. 2017, doi: [10.1007/s11081-016-9339-9](https://doi.org/10.1007/s11081-016-9339-9).
- [54] P. Das, J. Mathur, R. Bhakar, and A. Kanudia, "Implications of short-term renewable energy resource intermittency in long-term power system planning," *Energy Strategy Rev.*, vol. 22, pp. 1–15, Nov. 2018, doi: [10.1016/j.esr.2018.06.005](https://doi.org/10.1016/j.esr.2018.06.005).
- [55] D. U. Ziegler, G. Pretticco, C. Mateo, and T. G. San Roman, "Achieving active distribution networks through operational and architectural aspects analysis: A methodology for long-term planning," in *Proc. 26th Int. Conf. Exhib. Electr. Distrib.*, Sep. 2021, pp. 2679–2684, doi: [10.1049/icp.2021.1522](https://doi.org/10.1049/icp.2021.1522).
- [56] G. Mokryani, "Active distribution networks planning with integration of demand response," *Sol. Energy*, vol. 122, pp. 1362–1370, Dec. 2015, doi: [10.1016/j.solener.2015.10.052](https://doi.org/10.1016/j.solener.2015.10.052).
- [57] S. Wang, Z. Y. Dong, C. Chen, H. Fan, and F. Luo, "Expansion planning of active distribution networks with multiple distributed energy resources and EV sharing system," *IEEE Trans. Smart Grid*, vol. 11, no. 1, pp. 602–611, Jan. 2020, doi: [10.1109/TSG.2019.2926572](https://doi.org/10.1109/TSG.2019.2926572).
- [58] N. C. Koutsoukis, P. S. Georgilakis, and N. D. Hatzigargyriou, "Multistage coordinated planning of active distribution networks," *IEEE Trans. Power Syst.*, vol. 33, no. 1, pp. 32–44, Jan. 2018, doi: [10.1109/TPWRS.2017.2699696](https://doi.org/10.1109/TPWRS.2017.2699696).
- [59] S. Mohtashami, D. Pudjianto, and G. Strbac, "Strategic distribution network planning with smart grid technologies," *IEEE Trans. Smart Grid*, vol. 8, no. 6, pp. 2656–2664, Nov. 2017, doi: [10.1109/TSG.2016.2533421](https://doi.org/10.1109/TSG.2016.2533421).
- [60] European Parliament and Council of the European Union. (Jun. 14, 2019). *Directive (EU) 2019/944 of the European Parliament and of the Council of 5 June 2019 on Common Rules for the Internal Market for Electricity and Amending Directive 2012/27/EU (Recast)*. [Online]. Available: <https://eur-lex.europa.eu/eli/dir/2019/944/oj>
- [61] M. A. Mejia, L. H. Macedo, G. Munoz-Delgado, J. Contreras, and A. Padilha-Feltrin, "Medium-term planning of active distribution systems considering voltage-dependent loads, network reconfiguration, and CO₂ emissions," *Int. J. Electr. Power Energy Syst.*, vol. 135, Feb. 2022, Art. no. 107541, doi: [10.1016/j.ijepes.2021.107541](https://doi.org/10.1016/j.ijepes.2021.107541).
- [62] R. S. Pinto, C. Unsuhay-Vila, and F. H. Tabarro, "Reliability-constrained robust expansion planning of active distribution networks," *IET Gener., Transmiss. Distrib.*, vol. 16, no. 1, pp. 27–40, Jan. 2022, doi: [10.1049/gtd2.12263](https://doi.org/10.1049/gtd2.12263).
- [63] X. Zhang, W. Hua, Y. Liu, J. Duan, Z. Tang, and J. Liu, "Reinforcement learning for active distribution network planning based on Monte Carlo tree search," *Int. J. Electr. Power Energy Syst.*, vol. 138, Jun. 2022, Art. no. 107885, doi: [10.1016/j.ijepes.2021.107885](https://doi.org/10.1016/j.ijepes.2021.107885).

- [64] G. Celli, G. Pisano, S. Ruggeri, G. G. Soma, F. Pilo, C. Papa, C. Pregonoli, L. de Carolis, S. Ferrero, and F. Cazzato, "Distribution systems as catalysts for energy transition embedding flexibility in large-scale applications," *IEEE Access*, vol. 12, pp. 92227–92240, 2024, doi: [10.1109/ACCESS.2024.3421615](https://doi.org/10.1109/ACCESS.2024.3421615).
- [65] H. Laaksonen, H. Khajeh, C. Parthasarathy, M. Shafie-khah, and N. Hatziaargyriou, "Towards flexible distribution systems: Future adaptive management schemes," *Appl. Sci.*, vol. 11, no. 8, p. 3709, Apr. 2021, doi: [10.3390/app11083709](https://doi.org/10.3390/app11083709).
- [66] L. Strezoski and I. Stefani, "Utility DERMS for active management of emerging distribution grids with high penetration of renewable DERs," *Electronics*, vol. 10, no. 16, p. 2027, Aug. 2021, doi: [10.3390/electronics10162027](https://doi.org/10.3390/electronics10162027).
- [67] P. S. Georgilakis, "Review of computational intelligence methods for local energy markets at the power distribution level to facilitate the integration of distributed energy resources: State-of-the-art and future research," *Energies*, vol. 13, no. 1, p. 186, Jan. 2020, doi: [10.3390/en13010186](https://doi.org/10.3390/en13010186).
- [68] D. Sharma and P. Singh, "Optimal planning of distribute energy resources sizing and location problem—A review," in *Proc. 2nd Int. Conf. Inventive Res. Comput. Appl. (ICIRCA)*, Jul. 2020, pp. 500–504, doi: [10.1109/ICIRCA48905.2020.9182899](https://doi.org/10.1109/ICIRCA48905.2020.9182899).
- [69] A. S. N. Huda and R. Živanović, "Large-scale integration of distributed generation into distribution networks: Study objectives, review of models and computational tools," *Renew. Sustain. Energy Rev.*, vol. 76, pp. 974–988, Sep. 2017, doi: [10.1016/j.rser.2017.03.069](https://doi.org/10.1016/j.rser.2017.03.069).
- [70] I. Oladeji, P. Makolo, M. Abdillahi, J. Shi, and R. Zamora, "Security impacts assessment of active distribution network on the modern grid operation—A review," *Electronics*, vol. 10, no. 16, p. 2040, Aug. 2021, doi: [10.3390/electronics10162040](https://doi.org/10.3390/electronics10162040).
- [71] S. Repo, F. Ponci, D. D. Giustina, A. Alvarez, C. C. Garcia, Z. Al-Jassim, H. Amaris, and A. Kulmala, "The IDE4L project: Defining, designing, and demonstrating the ideal grid for all," *IEEE Power Energy Mag.*, vol. 15, no. 3, pp. 41–51, May 2017, doi: [10.1109/MPE.2017.2662329](https://doi.org/10.1109/MPE.2017.2662329).
- [72] H. Shishtawi, L. Grella, M. Jaggassar, F. Magnago, and N. Al-Hariri, "Implementation of the active network management scheme for the active response to distribution network constraints project," in *Proc. CIRED - 26th Int. Conf. Exhib. Electr. Distrib.*, vol. 2021, Sep. 2021, pp. 1644–1648, doi: [10.1049/icp.2021.1642](https://doi.org/10.1049/icp.2021.1642).
- [73] M. Wu, L. Kou, X. Hou, Y. Ji, B. Xu, and H. Gao, "A bi-level robust planning model for active distribution networks considering uncertainties of renewable energies," *Int. J. Electr. Power Energy Syst.*, vol. 105, pp. 814–822, Feb. 2019, doi: [10.1016/j.jepes.2018.09.032](https://doi.org/10.1016/j.jepes.2018.09.032).
- [74] N. C. Koutsoukis and P. S. Georgilakis, "A multistage distribution network planning method considering distributed generation active management and demand response," *IET Renew. Power Gener.*, vol. 16, no. 1, pp. 65–76, Jan. 2022, doi: [10.1049/rpg.2.12325](https://doi.org/10.1049/rpg.2.12325).
- [75] G. Mokryani, Y. F. Hu, P. Papadopoulos, T. Niknam, and J. Aghaei, "Deterministic approach for active distribution networks planning with high penetration of wind and solar power," *Renew. Energy*, vol. 113, pp. 942–951, Dec. 2017, doi: [10.1016/j.renene.2017.06.074](https://doi.org/10.1016/j.renene.2017.06.074).
- [76] Y. Li, Q. Chen, G. Strbac, K. Hur, and C. Kang, "Active distribution network expansion planning with dynamic thermal rating of underground cables and transformers," *IEEE Trans. Smart Grid*, vol. 15, no. 1, pp. 218–232, Jan. 2024, doi: [10.1109/TSG.2023.3266782](https://doi.org/10.1109/TSG.2023.3266782).
- [77] C. Pan, W. Chen, C. Zhou, and W. Zhou, "Distribution network planning considering active response of EVs and DTR of cables and transformers," *High Voltage*, vol. 10, no. 2, pp. 279–293, Apr. 2025, doi: [10.1049/hve.2.70024](https://doi.org/10.1049/hve.2.70024).
- [78] A. R. Jordehi, "Optimisation of electric distribution systems: A review," *Renew. Sustain. Energy Rev.*, vol. 51, pp. 1088–1100, Nov. 2015, doi: [10.1016/j.rser.2015.07.004](https://doi.org/10.1016/j.rser.2015.07.004).
- [79] M. K. Verma, V. Mukherjee, V. K. Yadav, and S. Ghosh, "Constraints for effective distribution network expansion planning: An ample review," *Int. J. Syst. Assurance Eng. Manage.*, vol. 11, no. 3, pp. 531–546, Jun. 2020, doi: [10.1007/s13198-020-00969-w](https://doi.org/10.1007/s13198-020-00969-w).
- [80] S. Mirjalili, "SCA: A sine cosine algorithm for solving optimization problems," *Knowl.-Based Syst.*, vol. 96, pp. 120–133, Mar. 2016, doi: [10.1016/j.knsys.2015.12.022](https://doi.org/10.1016/j.knsys.2015.12.022).
- [81] N. Saini and S. Saha, "Multi-objective optimization techniques: A survey of the state-of-the-art and applications," *Eur. Phys. J. Special Topics*, vol. 230, no. 10, pp. 2319–2335, Sep. 2021, doi: [10.1140/epjs/s11734-021-00206-w](https://doi.org/10.1140/epjs/s11734-021-00206-w).
- [82] B. Xin, L. Chen, J. Chen, H. Ishibuchi, K. Hirota, and B. Liu, "Interactive multiobjective optimization: A review of the state-of-the-art," *IEEE Access*, vol. 6, pp. 41256–41279, 2018, doi: [10.1109/ACCESS.2018.2856832](https://doi.org/10.1109/ACCESS.2018.2856832).
- [83] N. Gunantara, "A review of multi-objective optimization: Methods and its applications," *Cogent Eng.*, vol. 5, no. 1, Jan. 2018, Art. no. 1502242, doi: [10.1080/23311916.2018.1502242](https://doi.org/10.1080/23311916.2018.1502242).
- [84] J. Lu, J. Han, Y. Hu, and G. Zhang, "Multilevel decision-making: A survey," *Inf. Sci.*, vols. 346–347, pp. 463–487, Jun. 2016, doi: [10.1016/j.ins.2016.01.084](https://doi.org/10.1016/j.ins.2016.01.084).
- [85] A. Sinha, P. Malo, and K. Deb, "A review on bilevel optimization: From classical to evolutionary approaches and applications," *IEEE Trans. Evol. Comput.*, vol. 22, no. 2, pp. 276–295, Apr. 2018, doi: [10.1109/TEVC.2017.2712906](https://doi.org/10.1109/TEVC.2017.2712906).
- [86] T. Kleinert, M. Labbé, I. Ljubić, and M. Schmidt, "A survey on mixed-integer programming techniques in bilevel optimization," *EURO J. Comput. Optim.*, vol. 9, Jan. 2021, Art. no. 100007, doi: [10.1016/j.ejco.2021.100007](https://doi.org/10.1016/j.ejco.2021.100007).
- [87] Y. Beck, I. Ljubić, and M. Schmidt, "A survey on bilevel optimization under uncertainty," *Eur. J. Oper. Res.*, vol. 311, no. 2, pp. 401–426, Dec. 2023, doi: [10.1016/j.ejor.2023.01.008](https://doi.org/10.1016/j.ejor.2023.01.008).
- [88] M. Pourakbari-Kasmaei, M. Asensio, M. Lehtonen, and J. Contreras, "Trilateral planning model for integrated community energy systems and PV-based prosumers—A bilevel stochastic programming approach," *IEEE Trans. Power Syst.*, vol. 35, no. 1, pp. 346–361, Jan. 2020, doi: [10.1109/TPWRS.2019.2935840](https://doi.org/10.1109/TPWRS.2019.2935840).
- [89] R. Li, W. Wang, X. Wu, F. Tang, and Z. Chen, "Cooperative planning model of renewable energy sources and energy storage units in active distribution systems: A bi-level model and Pareto analysis," *Energy*, vol. 168, pp. 30–42, Feb. 2019, doi: [10.1016/j.energy.2018.11.069](https://doi.org/10.1016/j.energy.2018.11.069).
- [90] S. Solat, F. Aminifar, A. Safdarian, and H. Shayanfar, "An expansion planning model for strategic visioning of active distribution network in the presence of local electricity market," *IET Gener., Transmiss. Distrib.*, vol. 17, no. 24, pp. 5410–5429, Dec. 2023, doi: [10.1049/gtd.2.13053](https://doi.org/10.1049/gtd.2.13053).
- [91] R. Li, W. Wang, and M. Xia, "Cooperative planning of active distribution system with renewable energy sources and energy storage systems," *IEEE Access*, vol. 6, pp. 5916–5926, 2018, doi: [10.1109/ACCESS.2017.2785263](https://doi.org/10.1109/ACCESS.2017.2785263).
- [92] C. Huang, C. Wang, N. Xie, and Y. Wang, "Robust coordination expansion planning for active distribution network in deregulated retail power market," *IEEE Trans. Smart Grid*, vol. 11, no. 2, pp. 1476–1488, Mar. 2020, doi: [10.1109/TSG.2019.2938723](https://doi.org/10.1109/TSG.2019.2938723).
- [93] H. Zakernezhad, M. S. Nazari, M. Shafie-khah, and J. P. S. Catalão, "Multi-level optimization framework for resilient distribution system expansion planning with distributed energy resources," *Energy*, vol. 214, Jan. 2021, Art. no. 118807, doi: [10.1016/j.energy.2020.118807](https://doi.org/10.1016/j.energy.2020.118807).
- [94] V. B. Pamshetti, S. Singh, A. K. Thakur, S. P. Singh, T. S. Babu, N. Patnaik, and G. H. Krishna, "Cooperative operational planning model for distributed energy resources with soft open point in active distribution network," *IEEE Trans. Ind. Appl.*, vol. 59, no. 2, pp. 2140–2151, Mar. 2023, doi: [10.1109/TIA.2022.3223339](https://doi.org/10.1109/TIA.2022.3223339).
- [95] Y. Yao, Y. He, N. D. Hatziaargyriou, H. Wu, and P. Han, "A multistage expansion planning method for distribution networks incorporating an adaptive short-term correction mechanism," *Sustain. Energy Technol. Assessments*, vol. 83, Nov. 2025, Art. no. 104628, doi: [10.1016/j.seta.2025.104628](https://doi.org/10.1016/j.seta.2025.104628).
- [96] S. S. A. Kaabi, H. H. Zeineldin, and V. Khadkikar, "Planning active distribution networks considering multi-DG configurations," *IEEE Trans. Power Syst.*, vol. 29, no. 2, pp. 785–793, Mar. 2014, doi: [10.1109/TPWRS.2013.2282343](https://doi.org/10.1109/TPWRS.2013.2282343).
- [97] B. Zeng, J. Wen, J. Shi, J. Zhang, and Y. Zhang, "A multi-level approach to active distribution system planning for efficient renewable energy harvesting in a deregulated environment," *Energy*, vol. 96, pp. 614–624, Feb. 2016, doi: [10.1016/j.energy.2015.12.070](https://doi.org/10.1016/j.energy.2015.12.070).
- [98] M. Kabirifar, M. Fotuhi-Firuzabad, M. Moeini-Aghaie, and N. Pourghaderi, "Joint distributed generation and active distribution network expansion planning considering active management of network," in *Proc. 27th Iranian Conf. Electr. Eng. (ICEE)*, Apr. 2019, pp. 702–708, doi: [10.1109/IranianCEE.2019.8786665](https://doi.org/10.1109/IranianCEE.2019.8786665).
- [99] M. Kabirifar, M. Fotuhi-Firuzabad, M. Moeini-Aghaie, and N. Pourghaderi, "Multistage active distribution network integrated planning incorporating energy storage systems and active network management," in *Proc. IEEE 4th Int. Conf. Intell. Energy Power Syst. (IEPS)*, Sep. 2020, pp. 163–168, doi: [10.1109/IEPSS1250.2020.9263121](https://doi.org/10.1109/IEPSS1250.2020.9263121).

- [100] S. Xie, Z. Hu, L. Yang, and J. Wang, "Expansion planning of active distribution system considering multiple active network managements and the optimal load-shedding direction," *Int. J. Electr. Power Energy Syst.*, vol. 115, Feb. 2020, Art. no. 105451, doi: [10.1016/j.ijepes.2019.105451](#).
- [101] M. Moradipoor, M. P. Moghaddam, and M. R. Haghifam, "A flexible active distribution system expansion planning model: A risk-based approach," *Energy*, vol. 145, pp. 442–457, Feb. 2018, doi: [10.1016/j.energy.2017.12.160](#).
- [102] S. Najafi Ravadanegh, N. Jahanyari, A. Amini, and N. Taghizadehgan, "Smart distribution grid multistage expansion planning under load forecasting uncertainty," *IET Gener., Transmiss. Distrib.*, vol. 10, no. 5, pp. 1136–1144, Apr. 2016, doi: [10.1049/iet-gtd.2015.0673](#).
- [103] S. Behzadi and A. Bagheri, "A trade-off between line hardening and dynamic line rating by a new convex optimization model for resilient micro-grid-oriented expansion planning of reconfigurable smart distribution networks incorporated with renewable energy sources," *Energy*, vol. 8, Jul. 2025, Art. no. 100351, doi: [10.1016/j.nxener.2025.100351](#).
- [104] M. A. A. Al-Ja'afreh, B. Amjad, K. Rowe, G. Mokryani, and J. L. A. Marquez, "Optimal planning and forecasting of active distribution networks using a multi-stage deep learning based technique," *Energy Rep.*, vol. 10, pp. 686–705, Nov. 2023, doi: [10.1016/j.egyr.2023.07.014](#).
- [105] X. Shen, M. Shahidehpour, S. Zhu, Y. Han, and J. Zheng, "Multi-stage planning of active distribution networks considering the co-optimization of operation strategies," *IEEE Trans. Smart Grid*, vol. 9, no. 2, pp. 1425–1433, Mar. 2018, doi: [10.1109/TSG.2016.2591586](#).
- [106] M. Vukobratović, P. Marić, G. Horvat, Z. Balkić, and S. Sučić, "A survey on computational intelligence applications in distribution network optimization," *Electronics*, vol. 10, no. 11, p. 1247, May 2021, doi: [10.3390/electronics10111247](#).
- [107] J. Yumbra, J. M. Home-Ortiz, T. Pinto, J. P. S. Catalão, and J. R. S. Mantovani, "Optimal operational planning of distribution systems: A neighborhood search-based matheuristic approach," *Sustain. Energy, Grids Netw.*, vol. 38, Jun. 2024, Art. no. 101330, doi: [10.1016/j.segan.2024.101330](#).
- [108] A. Ehsan, Q. Yang, and M. Cheng, "A scenario-based robust investment planning model for multi-type distributed generation under uncertainties," *IET Gener., Transmiss. Distrib.*, vol. 12, no. 20, pp. 4426–4434, Nov. 2018, doi: [10.1049/iet-gtd.2018.5602](#).
- [109] N. Koutsoukis, P. Georgilakis, and N. Hatzigiorgiouri, "Active distribution network planning based on a hybrid genetic algorithm-nonlinear programming method," *CIREN-Open Access Proc. J.*, vol. 2017, no. 1, pp. 2065–2068, Oct. 2017, doi: [10.1049/oap-cired.2017.0771](#).
- [110] S. Kayalvizhi and D. M. V. Kumar, "Optimal planning of active distribution networks with hybrid distributed energy resources using grid-based multi-objective harmony search algorithm," *Appl. Soft Comput.*, vol. 67, pp. 387–398, Jun. 2018, doi: [10.1016/j.asoc.2018.03.009](#).
- [111] S. Xie, Z. Hu, D. Zhou, Y. Li, S. Kong, W. Lin, and Y. Zheng, "Multi-objective active distribution networks expansion planning by scenario-based stochastic programming considering uncertain and random weight of network," *Appl. Energy*, vol. 219, pp. 207–225, Jun. 2018, doi: [10.1016/j.apenergy.2018.03.023](#).
- [112] M. Kabirifar, M. Fotuhi-Firuzabad, N. Pourghaderi, P. Dehghanian, M. Lehtonen, and F. Wang, "Expansion planning in restructured active power distribution networks considering agents' investment budget uncertainty," in *Proc. 4th Int. Conf. Smart Grid Renew. Energy (SGRE)*, Jan. 2024, pp. 1–6, doi: [10.1109/sgre59715.2024.10429009](#).
- [113] Y. Wang, X. Shen, and Y. Xu, "Joint planning of active distribution network and EV charging stations considering vehicle-to-grid functionality and reactive power support," *CSEE J. Power Energy Syst.*, vol. 10, no. 5, pp. 2100–2113, Sep. 2024, doi: [10.1775/CSEEJPES.2023.03930](#).
- [114] J. Li, F. Li, K. Sun, and B. Sun, "Bi-level coordinated expansion planning of high-PV-penetration active distribution networks incorporating multiple flexible resources considering WGAN-GP-based uncertainty generation," *Energy*, vol. 330, Sep. 2025, Art. no. 136685, doi: [10.1016/j.energy.2025.136685](#).
- [115] C. Zhang, Y. Ding, Q. Wu, Q. Wang, and J. Østergaard, "Distribution network expansion planning based on multi-objective PSO algorithm," *Energy Power Eng.*, vol. 5, no. 4, pp. 975–979, Jul. 2013, doi: [10.4236/epe.2013.54b187](#).
- [116] M. Ovaere, E. Heylen, S. Proost, G. Deconinck, and D. Van Hertem, "How detailed value of lost load data impact power system reliability decisions," *Energy Policy*, vol. 132, pp. 1064–1075, Sep. 2019, doi: [10.1016/j.enpol.2019.06.058](#).
- [117] M. Jooshaki, M. Lehtonen, M. Fotuhi-Firuzabad, G. Muñoz-Delgado, J. Contreras, and J. M. Arroyo, "On the explicit formulation of reliability assessment of distribution systems with unknown network topology: Incorporation of DG, switching interruptions, and customer-interruption quantification," *Appl. Energy*, vol. 324, Oct. 2022, Art. no. 119655, doi: [10.1016/j.apenergy.2022.119655](#).
- [118] A. Escalera, B. Hayes, and M. Prodanović, "A survey of reliability assessment techniques for modern distribution networks," *Renew. Sustain. Energy Rev.*, vol. 91, pp. 344–357, Aug. 2018, doi: [10.1016/j.rser.2018.02.031](#).
- [119] D. Enescu, P. Colella, A. Russo, R. F. Porumb, and G. C. Seritan, "Concepts and methods to assess the dynamic thermal rating of underground power cables," *Energies*, vol. 14, no. 9, p. 2591, May 2021, doi: [10.3390/en14092591](#).
- [120] C.-M. Lai and J. Teh, "Comprehensive review of the dynamic thermal rating system for sustainable electrical power systems," *Energy Rep.*, vol. 8, pp. 3263–3288, Nov. 2022, doi: [10.1016/j.egyr.2022.02.085](#).
- [121] R. Peña, A. Colmenar-Santos, and E. Rosales-Asensio, "Dynamic line rating: Technology and future perspectives," *Electronics*, vol. 14, no. 14, p. 2828, Jul. 2025, doi: [10.3390/electronics14142828](#).
- [122] S. M. Abdelkader, S. Kinga, E. Ebinyu, J. Amissah, G. Mugerwa, I. B. M. Taha, and D.-E.-A. Mansour, "Advancements in data-driven voltage control in active distribution networks: A comprehensive review," *Results Eng.*, vol. 23, Sep. 2024, Art. no. 102741, doi: [10.1016/j.rineng.2024.102741](#).
- [123] M. Atta and M. F. Nadeem Khan, "Voltage control in electric vehicles integrated active distribution networks—A review," in *Proc. 21st Int. Bhurban Conf. Appl. Sci. Technol. (IBCAST)*, Aug. 2024, pp. 630–635, doi: [10.1109/ibcast61650.2024.10877237](#).
- [124] W. Murray, M. Adonis, and A. Raji, "Voltage control in future electrical distribution networks," *Renew. Sustain. Energy Rev.*, vol. 146, Aug. 2021, Art. no. 111100, doi: [10.1016/j.rser.2021.111100](#).
- [125] M. N. I. Sarkar, L. G. Meegahapola, and M. Datta, "Reactive power management in renewable rich power grids: A review of grid-codes, renewable generators, support devices, control strategies and optimization algorithms," *IEEE Access*, vol. 6, pp. 41458–41489, 2018, doi: [10.1109/ACCESS.2018.2838563](#).
- [126] C. Kumar, M. Lakshmanan, S. Jaisiva, K. Prabaakaran, S. Barua, and H. H. Fayek, "Reactive power control in renewable rich power grids: A literature review," *IET Renew. Power Gener.*, vol. 17, no. 5, pp. 1303–1327, Apr. 2023, doi: [10.1049/rpg2.12674](#).
- [127] H. H. H. Mousa, K. Mahmoud, and M. Lehtonen, "A comprehensive review on recent developments of hosting capacity estimation and optimization for active distribution networks," *IEEE Access*, vol. 12, pp. 18545–18593, 2024, doi: [10.1109/ACCESS.2024.3359431](#).
- [128] Z. M. Zenhom, S. H. E. A. Aleem, A. F. Zobaa, and T. A. Boghdady, "A comprehensive review of renewables and electric vehicles hosting capacity in active distribution networks," *IEEE Access*, vol. 12, pp. 3672–3699, 2024, doi: [10.1109/ACCESS.2023.3349235](#).
- [129] S. Karagiannopoulos, P. Aristidou, A. Ulbig, S. Koch, and G. Hug, "Optimal planning of distribution grids considering active power curtailment and reactive power control," in *Proc. IEEE Power Energy Soc. Gen. Meeting (PESGM)*, Jul. 2016, p. 1, doi: [10.1109/PESGM.2016.7741538](#).
- [130] B. Yang, R. Zhang, J. Zhang, X. Cheng, J. Li, Y. Zhou, Y. Hu, B. He, G. Zhang, X. Du, S. Ji, Y. Zhang, and D. Guo, "A critical review of active distribution network reconfiguration: Concepts, development, and perspectives," *Energy Eng.*, vol. 121, no. 12, pp. 3487–3547, 2024, doi: [10.32604/ee.2024.054662](#).
- [131] B. Vena and K. Reddy, "Distribution network reconfiguration via metaheuristic optimization techniques," in *Proc. 33rd Southern Afr. Universities Power Eng. Conf. (SAUPEC)*, Jan. 2025, pp. 1–6, doi: [10.1109/saupec65723.2025.10944418](#).
- [132] W. Fadel, "A review of distributed generation optimization with shunt capacitors in reconfigured distribution networks," *Wireless Pers. Commun.*, vol. 136, no. 1, pp. 1–33, May 2024, doi: [10.1007/s11277-024-11062-x](#).
- [133] E. H. Sadiq and R. K. Antar, "Minimizing power losses in distribution networks: A comprehensive review," *Chin. J. Electr. Eng.*, vol. 10, no. 4, pp. 20–36, Dec. 2024, doi: [10.23919/cjee.2024.000093](#).



of Electrical Engineering, Tampere University. Her research interests include the integration of distributed energy resources into electricity distribution networks, active network management, and the planning of active distribution networks.



the whole energy system. The focus of research is on application of ICT and distribution automation to integrate distributed energy resources for network management. Recently the market-based flexibility utilization for the distribution grid congestion management has been the key objective of the research.

IRINA LAAMANEN was born in Santiago de Cuba, Cuba, in 1986. She received the M.Sc. degree in electrical engineering from the University of Oriente, Cuba, in 2015. She is currently pursuing the Ph.D. degree (Tech) in computing and electrical engineering with Tampere University, Finland. From 2015 to 2019, she held a teaching position in the Faculty of Electrical Engineering with the University of Oriente. Since 2021, she has been a Doctoral Researcher with the Department

SAMI REPO was born in Simpele, Finland, in 1969. He received the M.Sc. and Ph.D. degrees in electrical engineering from Tampere University of Technology, Tampere, Finland, in 1996 and 2001, respectively.

He is currently a Professor with the Unit of Electrical Engineering, Tampere University. His main research interests include management of active distribution networks including distributed energy resources and large-scale electrification of



THOMAS STETZ was born in Gelnhausen, Germany, in 1983. He received the Diploma degree in industrial engineering (Dipl.-Wi.-Ing. (FH)) from the University of Applied Sciences Darmstadt, Germany, in 2008, and an M.S. degree in renewable energies and energy efficiency from the University of Kassel, Germany, in 2009. He received the Ph.D. degree in electrical engineering from the University of Kassel, in 2014.

From 2009 to 2013, he was with the Fraunhofer IWES (now Fraunhofer IEE) as a Research Assistant. From 2013 to 2015, he served as the Head of the Research Group “Operation and Planning/Hybrid Grids.” Since 2015, he has been a Professor of Smart Grids and Energy Storage with the Department of Electrical Engineering and Information Technology, THM University of Applied Sciences, Giessen, Germany. Since 2023, he has also served as a Scientific Advisor to ITEE e.V. (Institute for Transformation Tasks in Energy Economics and Energy Technology), Giessen. His research interests include the optimized planning and operation of electric distribution systems. He is the authored or co-authored of more than 40 scientific publications.

Prof. Stetz is a member of German Association for Electrical, Electronic and Information Technology.

• • •